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A Preliminary Workflow of Fast Probabilistic Geomechanics for Containment
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Executive summary

The properties of geomaterials vary within a considerable range. If the mean or characteristic values of the parameters are used in deterministic analysis, the underlying uncertainties of the parametric values cannot be explicitly reflected in the final response. As the geomechanical model becomes increasing complex, the number of uncertainties increases accordingly. Consequently, the complexities of the interplay between the factors and the responses increases in an exponential fashion. At a tipping junction, it becomes difficult for one to distinguish and rank in a quantitative manner the combinatoric effects of each uncertainties that contribute to a specific response.

In order to predict a stochastic system responses, two serials tactics were proposed in this report to deal with; firstly the complexities of the inter-woven relationships between the state variables and its responses, and secondly probabilistic approach to capture the parameters variability. The first tactic, named the screening unit, aims to unravel the complexities of the input parameters and its responses. In other words, it seeks to sieve out the important parameters from the potentially important parameters. The screening is achieved via the Design of Experiments and Multi-Variate Regression. Once the thread of these complexities has been disentangled, probabilistic method took over. Using a combination of First-Order Reliability Method and Monte Carlo, the design point of the stochastic system is located and the corresponding fragility curve drawn for reliability assessment.

A spreadsheet implementation of the Fast Probabilistic Workflow is prototyped via a Geertsma subsidence problem where analytical solution is known. This demonstration intended to provide a preliminary proof of concept on probabilistic computation and generating a workflow template of sequence to approach geomechanics uncertainty in a systematic way.

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1. Introduction

In reservoir geomechanics, one is continually required to predict the mechanical response of a reservoir with an underlying hosts of uncertainties as input – a procedure called *de-risking*.

These uncertainties may include such variation in the operating pressure, depth and thickness of the reservoir, angle of dip, OWC/GOC, earth stress ratios and orientation, intensity of fracturing, mechanical and failure properties of each geological strata etc., to name a few. The geomechanics response, on the other hand, may include surface displacement, stress arching, reservoir compaction, fault reactivation and stress intensity around wellbore, by no means exhaustive. As the number of uncertainties in the geomechanics model increases, the complexities of the interplay between the factors and the responses increases exponentially. Under this situation, it is difficult for one to distinguish and rank in a quantitative manner the combinatoric effects of each uncertainties that contribute to a specific response.

1.1. Probabilistic Geomechanics

1.1.1. Uncertainties screening

The ability to associate uncertainties in the input to a specific response allows one to *seive out* the most important uncertainties from a 'basketfull' of potentially important one. This has an immediate positive results in project's cost and time saving because it helps the operator to set the bottomline by screening out *realistically* important and *potentially* important uncertainties. By focusing on the real important uncertainties of uncertainties in the input, the risk bandwidth in a system output is thus narrowed. The ranking of the importance of each uncertainties in relations to their impact on the response are crucial in reservoir management because it hints to the operator *how* and *where* to allocate their geomechanics resources by means of cost/time adjustment in surveillance data, laboratory experiments and numerical modeling.

1.1.2. Reliability Assessment

Screening by itself only sieves out the important uncertainties and reduces the associated risk bandwidth; it does not inform us how the uncertainties contribute to failure nor does it inform us how reliable is the safety prediction. In the past, engineers like to use the concept of safety factor, determined by

$$\text{safety factor} = \frac{\text{average strength}}{\text{worst expected stress}}, \quad \{1\}$$

to indicate the integrity of their design. The problem with equation (1) is that it sends out a wrong signal that the higher is the safety factor, the safer is the design. Unfortunately, this approach fails times and again throughout history [10] because it masks the variability in strength and stress of the design, consequently underestimating the hidden risk. The same logic applies to reservoir geomechanics.

The properties of geomaterials are stochastics in nature and vary within a considerable range. If the mean or characteristic values of the parameters are used in deterministic analysis, the underlying uncertainties of the parametric values cannot be explicitly reflected in the final response. To overcome this inherent variability, the probabilistic values of the input parameters need be to taken into account. This is achieved in this project by using the First-Order Reliability Method.

By combining the screening tool with reliability analysis, one offers a straightforward probabilistic geomechanics workflow that assist field engineers to make design decision such as allowable injection reservoir pressure. Critical key challenges such as the allowable range of safe depleting pressure and the associated likelihood of mechanical failures can be estimated in a quantitative and systematic manner.

1.2. Objectives of the Project

There are two primary objectives for the current project, summarily enumerated below:

- i. Propose a straightforward *fit-for-purpose probabilistic geomechanics workflow* that combines the uncertainty screening and safety reliability assessment.
- ii. Demonstrate that the proposed workflow is *feasible* and *workable* in real time.

It is important to highlight that the following aspects as spin-offs from the objectives above where the results fit in a grandeur picture:

- a. Establish a probabilistic footprint in *Geomechanics 2020 Vision*, updating the practice from a more traditional deterministic analysis towards probabilistic methodologies
- b. Deliver a simple and speedy geomechanics assessment workflow for Shell ORP, which focuses on answering key geomechanics questions. The current project will fit comfortably in Shell's ORP Phrase I as shown in Fig. 1. It is envisions that in the mid-term, the project will be extended to cover the ORP Select Phase when more elaborated probabilistic functionalities are added. Ultimately, in the longer term, the goal is to close the geomechanics loop in the ORP Operate Phase where surveillance data are incorporated into the probabilistic

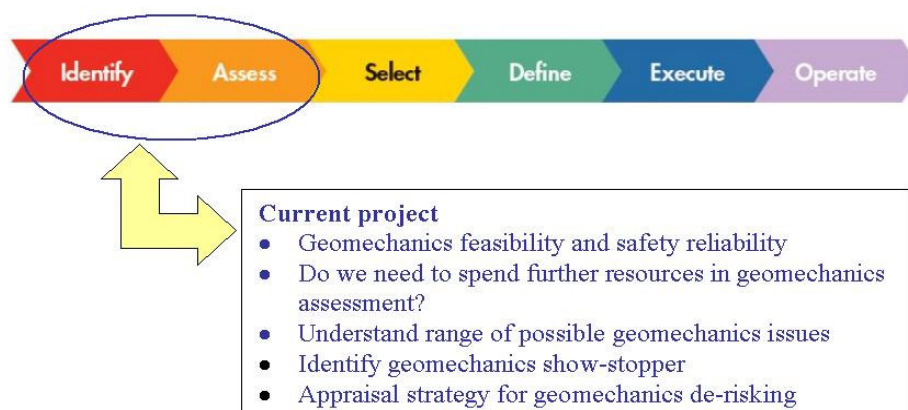


Figure 1: Fast probabilistic geomechanics within Phase I of Shell ORP

- c. Establish a general workflow for *component reliability* calculations, leading to the ultimate formulation of geomechanics system reliability in the Reliacon Project, see Fig. 2. Due to the timeline/resources limitation, this report will concentrate only on one component reliability calculation for demonstration purpose.

- d. Enact the 'bone and meat' of the *geomechanics safety assessment* for the high level management decision making tool, *TESLA*, used by the CCS and Sour gas projects, see Fig. 2.

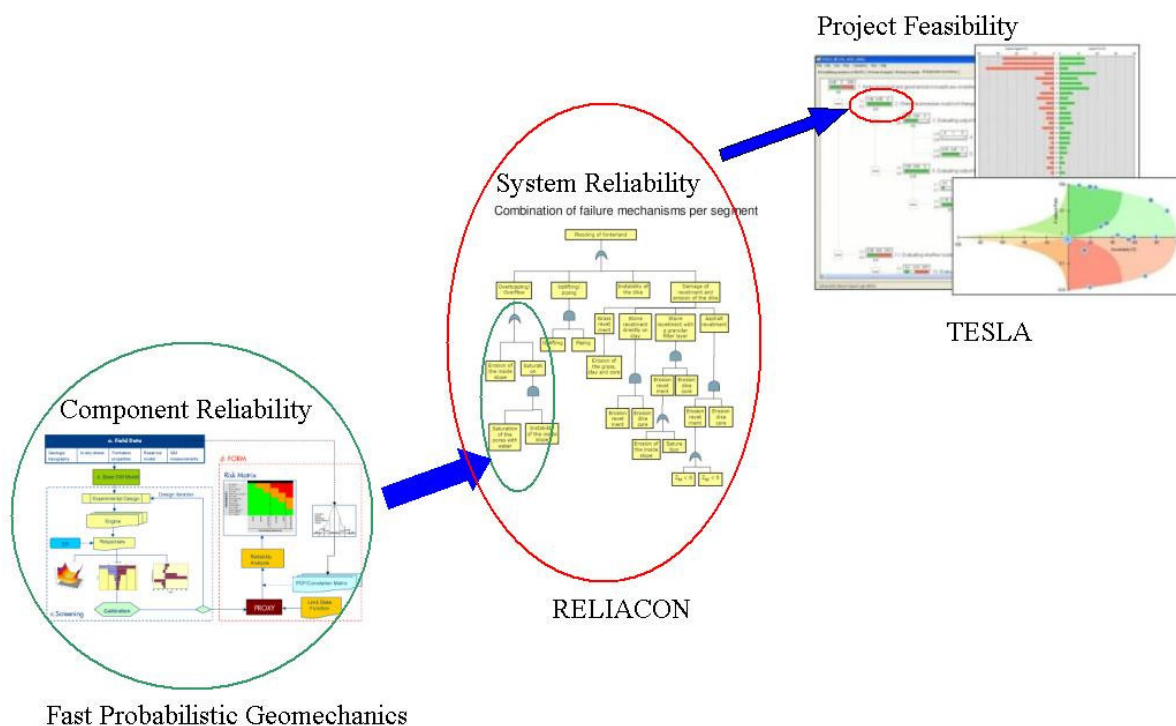


Figure 2: Relative positioning of the fast probabilistic geomechanics with respect to other projects in Shell

1.3. Scope of the Project

The aforementioned uncertainty screening with reliability analysis forms the two core deliverables of this report. Uncertainty screening comprises of two main components, namely Design of Experiments (DoE) and Multi-Variate Regression (MVR) while the reliability analysis constitutes the use of First-Order Reliability Method (FORM) and simple Monte Carlo method.

DoE and MVR are old and well-established subjects with wide-ranging applications in manufacturing, social science and bio-medical industries. It is not the purpose of this research to further any investigation of the methods. Attention is focused solely on fit-for-purpose application of DoE and MVR in geomechanics. As DoE and MVR has never been used before in geomechanics in such a manner, the emphasis is placed on the feasibility of the methodology in geomechanics for EP application.

On the other hand, geomechanics is a very broad and complex subject with an unique discipline by its own right. It is impossible to tackle the geomechanics subject in one single R&D development project. It is thus rational to set the limitation and scope of this project by means of the resources and timeline availability. Rather than venturing into different type of probabilistic methods with brute force (e.g. Monte Carlo finite element simulation), attention is focused on fit-for-purpose semi-analytical probabilistic method, which has a clear geometric representation in its definition of safe reliability index.

Apart from the above two constraints, it is important to spell out here that this project *do not* intend to investigate theoretical, computational and software implementational aspects of geomechanics. The geomechanics tool is thus treated as a 'black-box' engine that spits out desirable responses at specified locations.

The scope of this project can be summarized as follows:

1. A brief overview of Design of Experiment (DoE) and the Multi-Variate Regression (MVR) to provide background information needed as the stepping-stone for the geomechanics screening unit.
2. A systematic way to extract the 'proxy' or the approximating functions representative of the uncertainties in the geomechanics model. As a first step in this project, it will be limited to first-order approximation.
3. Using geomechanics tool to investigate the response of a specified type of failure mechanism, in this case, surface subsidence due to production. The reason for choosing this mechanism is because of its simplicity in identifying the failure and availability of analytical solution.
4. Institute a set of workable failure criteria, ranging from the very conservative to a very 'progressive' one. This is effect to investigate the implication of failure criterion on the reliability index.
5. Establish a systematic way to quantify the impact of each uncertainties and a ranking system for the sensitivity of each.
6. Estimating the upper and lower error bound of the approximating function.
7. Perform safety reliability assessment based on the approximating function and identifying the critical states.

1.4. Probabilistic Geomechanics on HSE and Containment

Probabilistic geomechanics has substantial importance on HS&E.

- **Top and bottom caprock integrity.** Probabilistic geomechanics modelling allows the uncertainty of the mechanical strength of the rocks forming the natural geological seals that capture the hydrocarbon to be analysed in an in-depth fashion for containment purpose. Reliable reservoir operational design can then be optimise to maximise the recovery while maintaining the structural and flow integrity. Shell's future business venture into unconventional resources requires extensive geomechanics input to ensure that production is environmentally safe. Carbon Capture and Storage (CCS) is also an increasing important area where caprock integrity and geomechanical safety plays an increasing role.
- **Geologic fault seal integrity.** A fault can be a transmitter of or barrier to fluid flow and pressure communication during the lifetime of an underground CO₂ storage. Categorising and estimating fault behaviour within these extremes is important for containment assessment. Key to this fault seal integrity issue is the change of the effective stresses around the faults when the CO₂ is injected. Areas in the vicinity of the top of a reservoir is prompted to stress arching effect and is most likely to transform a sealing fault into a fluid transmitter.
- **Fault reactivation.** A change in fluid pressure within a porous reservoir will induce stress changes within the reservoir and the rocks that surround it. Amongst the potential hazards resulting from these induced changes is the reactivation of existing faults, which may breach the hydraulic integrity of the caprocks that bound the reservoir.
- **Surface and subsurface alteration.** Surface compaction may result from hydrocarbon production in structurally weak formation while heaving may occur in the case of carbon

injection. These mechanisms can lead to severe surface facilities distortion due to surface topological changes. Sub-surface wellbore issues may also be encountered due to bucking of wellbore and disruption of well productivity. Probabilistic method can prevent these undesirable surface and subsurface alteration by minimising the effective stress changes in the overburden with regards to the production.

- **Wellbore damage.** In enhanced recovery and well stimulation, heat is injected into the reservoir rock to improve recovery. Geomechanically, this will cause it to expand in proportion to the temperature increases. Thermal expansion of borehole will result in squeezing effect of the hole wall against the well casing, eventually crushing it. The removal of mass due to pyrolysis in Shell's unconventional oil operation, for example, will generate a dynamic compaction, where the well may be squashed vertically, causing it to buckle. With the help of reliability method, a suitable well path can be designed and cautionary measures can be taken to prevent well loss.

1.5. Report Layout

Chapter 2 gives a brief overview of probabilistic geomechanics methods within the context of deterministic analysis, introducing some fundamental concepts and fit-for-purpose probabilistic application. An illustrated workflow diagram is used to demonstrate how the current probabilistic analysis can be inserted into existing deterministic work practices.

In Chapter 3, the feasibility of the workflow is tested using a geomechanics model. A semi-analytical Geertsma model, implemented in Excel spreadsheet, is used as a geomechanics 'black-box' to provide the necessary deterministic calculation. The geomechanics model used was greatly simplified in its complexity so as the problem itself does not overwhelm the report's objectives.

In Chapter 4, we conclude the report and recommend steps to take the project forward and further.

Appendix 1 supplemented this report with technical background of the Experimental Design and Response Surface Analysis. Appendix 2 provided the literature survey of probabilistic methods in geomechanics analysis. Appendix 3 listed the available options for software selection; pros and cons of each software and final decision.

2. Fast Probabilistic Geomechanics Workflow

The objectives of this section is to provide a brief overview of the background involved in Probabilistic Geomechanics which in turn involved Design of Experiment (DoE), Multi-Variate Regression (MVR); and Safety Reliability Assessment. A few key concepts are now introduced below. For detailed description of each concept, a good starting point is Ref [4].

2.1. Types of Uncertainty

To set the stage, the term uncertainty needs to be defined. The following three types of uncertainties can be identified [1] :

- i) *Inherent uncertainty* due to variability of input and/or model parameters when the characterisation of the variability is not available (e.g. irrational behaviour in the stock market)
- ii) *Uncertainty* due to variability of input and/or model parameters when the characterisation is available (e.g. variability of geomaterial with probability density functions)
- iii) *Knowledge/model uncertainty*, in other words, uncertainty due to an unknown process or mechanism.

Type (i) uncertainty, which depends on chance, may be referred to as statistical uncertainty. Statistical uncertainty is irreducible. Type (ii) and (iii) are collectively referred to as systematic uncertainties because they are reducible to some extent. Type (ii) uncertainty can be reduced by taking more measurements while type (iii) uncertainty can be reduced by doing research.

It often happens in real life application that all three types of uncertainties are present in the system under study. Uncertainty quantification intends to work towards reducing Type (i) and (iii) to type (ii). In the context of the present project, the proposed methodology for geomechanics helps to reduce the type (i) and (iii) uncertainties to input parameters uncertainties.

2.2. Safety Factor and Reliability

Reliability of a system can be defined as the ability of a system or component to perform its required functions under a specified conditions [2]. It is often reported as a probability. In the past, engineers like to use the concept of safety factor [3], determined by

$$\text{safety factor} = \frac{\text{average strength}}{\text{worst expected stress}}, \quad \{2\}$$

to indicate the integrity of their design. Typically, a safety factor below unity is considered failure. The problem with equation (2) is that it sends out a wrong signal that the higher is the safety factor, the safer is the design. This approach fails times and again because it masks the variability in strength and stress of the design, consequently underestimating the hidden risk. To estimate the real safety factor, variability of the parameters must be dealt with explicitly.

In order to frame the concept of reliability in relation to the safety factor according to Eq. (2), Fig. 1 shows the probabilistic density distribution of the average strength and worst expected stress for two designs based on their characteristic values. The blue curve represents the worst expected applied load with a mean value of 5 GPa and red curve that of the average strength of the material with a mean value of 8.5 GPa. Both systems in Fig. 3a and Fig. 3b have the same safety factor, $S=1.7$ according to Eq. (2).

The key question now is: how reliable is $S=1.7$ for Fig.3a and Fig.3b?

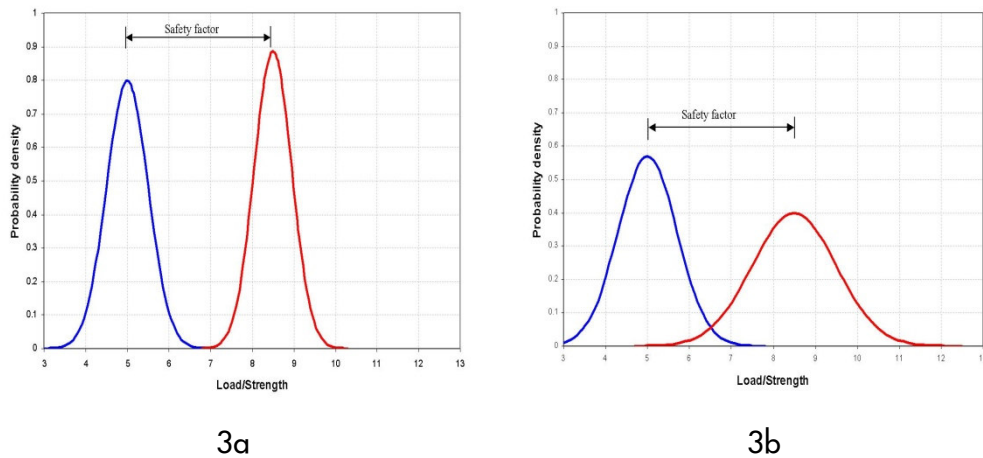


Figure 3: Safety factor reliability, blue curve is the worse expected stress and red curve is the average strength of the material. (a) No conflict between stress and strength; (b) the situation may occur that the stress (loading) exceeds the strength although the safety factor is the same.

Referring to Fig.3a, at the extreme cases (where the two curves touch), the minimum possible safety factor is approximately unity – which is still safe. This is not the situation for Fig. 3b. At the extreme case, $S \sim 0.625$, a value far below the safety limit of unity. We say system Fig. 3a is less likely to fail than system Fig.3b, i.e. more reliable. Intuitively, this points to the definition of reliability – *it is an indicator of how far away from the worst-case scenario*. A more precise definition will be given later in this Chapter. It should be clear from this example that 'safety factor' alone is inadequate to uncover the hidden risk. The same principle applies to reservoir geomechanics.

If the mean or characteristic values of the input are used in deterministic geomechanics analysis, the underlying variability of the parametric values cannot be explicitly reflected in the final response. It is not the characteristic values of the parameters that hide the risk, but the variability of the parameters. This variability is represented statistically by the standard deviation, σ . Hence, the higher the σ , the more uncertainty there is in a parameter. Figure 4 shows a standard normal Bell-shaped curve of a probabilistic density function and its correlation to standard deviation and cumulative percentiles. In Shell, the standard practice of using the cumulative P_{10} , P_{50} and P_{90} values correspond to -1.28σ , 0 , 1.28σ in a normally distributed probabilistic curve, respectively.

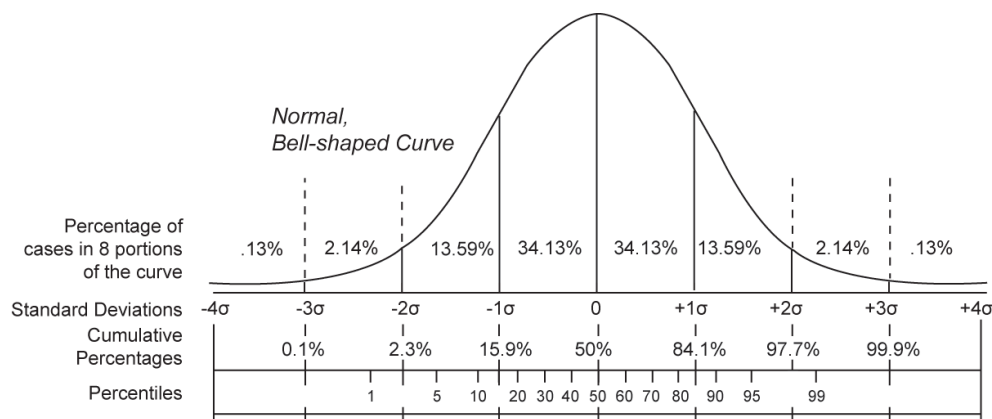


Figure 4: Standard normal Bell-shaped curve and the associated standard deviations and cumulative percentiles

2.3. Design of Experiment

According to the Oxford Dictionary, an experiment is “a test under controlled conditions that is made to ... examine the validity of a hypothesis...”. Design of experiments [4] is the design of an *information-gathering-screening* exercise where variation is present, whether under the full control of the experimenter or not. Often the experimenter is interested in the response of some processes of a particular system. It is thus a discipline that has very broad application.

Paramount to the goal of DoE is the designs used for *screening*. For example, they can be performed to gain insight into which factors are important in a particular safety issue in a given production/injection scenario. Consequently, screening allows one to narrow down the long list of so-called ‘potentially important’ factors into those that are *really* important with a known amount of statistical confidence.

The procedure by which one would intuitively accomplish a screening design is by running a given factor at two extreme levels (e.g. high-low, min-max etc) and record the response. The simplest design for accomplishing this is called the two-level full factorial design. This involves a brute force approach where every factor is run against all the other factors at all their possible settings in an exhaustive way. The immediate short coming of this design is that the number of experiments can be huge. It is *exponentially* proportional to the number of factors according to 2^n . For example, $2^6=64$ runs is for a two-level, full factorial design with six factors. Consequently, one can quickly run out of resources in experimenting with only a modest number of factors. This is referred to as full factorial design.

2.3.1. Fractional factorial and Plackett-Burman design

The basic purpose of a fractional factorial design is to overcome the shortcoming of full factorial design. It economically investigates cause-and-effect relationships of significance in a given experimental setting. An important property of a fractional design is its resolution or ability to separate main effects and low-order interactions from one another. The most important fractional designs are those of resolution III, IV and V: Resolutions below III are not useful because main effects are confounded with one another while resolutions above V are wasteful in that they estimate very high-order interactions which may be unphysical in practice. Within the current context, the research is limited to designs of resolution III, whose main objective is to sieve out the

main effects from the potentials. For this reason, these designs are often termed main effects designs, or *screening designs*.

To understand the fundamental idea of fractional factorial design, consider the two-level factorial design for three factors, namely the 2^3 design, represented by the cube in Fig. 1. X_1 , X_2 , X_3 represents 3 uncertainty factors, and the circles numbered '1' through '8' represent the responses from the experiments.

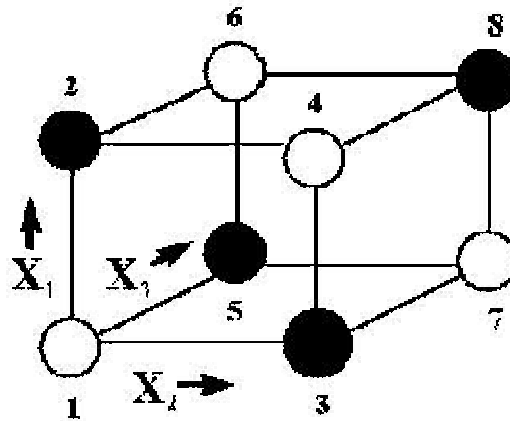


Figure 5: A 2^3 Full Factorial Design; Factors X_1 , X_2 , X_3 . (The arrows show the direction of increase of the factors. Numbers '1' through '8' at the corners of the design cube reference the 'Standard Order' of runs). Ref. [4]

It could be shown mathematically that if the responses from the white circles are known, the responses from the shaded circles can be determined uniquely, and vice versa. In other words, it is sufficient to run 4 experiments instead of $2^3=8$ experiments in order to determine the full responses due to uncertainties in X_1 , X_2 , X_3 .

Plackett-Burman design [4] is a special class of fractional factorial design. It allows construction of experiments with the run number a multiple of four. Experiments containing up to 27 uncertain factors in a mere 28-run is possible with Plackett-Burman design. Because of its efficiency, Plackett-Burman design will be used throughout this report. A more elaborated description of fractional factorial design can be found in Appendix.

2.3.2. Limitations of fractional factorial experiments

Before the two-level DoE can be used, its limitation must be understood. Because it requires only two extreme values from each state variables to predict the system responses, interactions between state variables and second-order effects cannot be seen from these experiments. Only the first-order effects can be seen in the two-level screening design. State variables chosen for a two-level screening must be independent of each other. For example, in-situ vertical stress, S_v , cannot be chosen together with horizontal stress ratio, $K_h=S_h/S_v$ because the value of K_h depends on S_v . However, choosing S_v and S_h together is a valid combination because they are independent of each other.

2.4. Multi-Variate Regression

Multi-Variate Regression (MVR) is a method used to explore the relationships between several exploratory variables and one or more response variables [4]. The method was introduced by G.E.P. Box and K.B. Wilson in 1951. It is widely acknowledged that MVR is a very crude approximation. However, this does not stop its popularity in a wide field of sciences and industries. This is partly because it is easy to apply and simple to understand. After all, the greatest strength about MVR is when little is known about the process, it provides a fit-for-purpose response estimates when experimental resources are scarce.

The core process of MVR consists of exploring the space of exploratory variables to develop an approximating relationship between the system response and the variables. This approximation is called a 'proxy' or more commonly known as *response surface*, i.e. an approximating model between the response Y and uncertain variables $x_1, x_2, \dots, x_n$, where n is the number of uncertainties. In general, the relationship between a response and its processes can be written as

$$Y = g(x_1, x_2, \dots, x_n) + err \quad \{3\}$$

where g is unknown and err is an error term that represents other sources of variability not accounted for. The error includes effects such as measurement error on the response, background noise, the effect of other variables and is treated as a statistical error.

The variables $x_1, x_2, \dots, x_n$ are called the natural variables, because they are expressed in natural units of measurement, such as degrees Celsius, MPa etc. It is sometimes more convenient to transform the natural variables to coded variables $X_1, X_2, \dots, X_n$, which are defined to be dimensionless with zero mean and plus-minus unity for the maximum-minimum values, respectively. In terms of the coded variables, the response function equation {3} becomes

$$Y = f(X_1, X_2, \dots, X_n) + err(\mathbf{X}) \quad \{4\}$$

The reason coded variables is preferred over natural variables is to minimize numerical ill conditioning during the algebraic manipulation. In addition, there are many situations where natural variable is an unnatural choice, e.g. reservoir pressure profile. In the situation when coded-variable is not possible, the magnitude difference between any two state variables should be limited to below 1×10^6 , as rule of thumb.

To further process equation {4}, linear regression can be used. Written in vector form, equation {4} appears as

$$\mathbf{Y} = \mathbf{X}\mathbf{A} + \boldsymbol{\varepsilon} \quad \{5\}$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} X_{11} & \cdots & X_{1p} \\ X_{21} & \cdots & X_{2p} \\ \vdots & \ddots & \vdots \\ X_{n1} & \cdots & X_{np} \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} A_1 \\ A_2 \\ \vdots \\ A_p \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_p \end{pmatrix} \quad \{6\}$$

The quantity $\mathbf{Y}$ is the observation, $\mathbf{X}$ is the design matrix, $\mathbf{A}$ is called the effects vector, and $\boldsymbol{\varepsilon}$ is the error. The main purpose is to find the solution to $\mathbf{A}$ by minimize the least squared error. Thus, we seek to compute

$$\mathbf{A} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \quad \{7\}$$

There are many methods available in the open literature [11, Chap 5] on solving equation {7} and will not be further discussed herein.

2.4.1. Limitation of the MVR

Geomechanical behavior is never linear. Even when the linear elastic assumption is made, there is no guarantee that a geomechanics model will behave linearly. This is because a geological model consists of strata of different thickness of different materials with wide range of differences in their mechanical properties. Added to these complexities is the present of fault lines, compartmentalization of reservoirs and aquifers, different magnitude of prehistoric stress conditions and some other unknown factors. Thus equation {5} must be strictly treated as a first-order approximation. The more complex the mathematical representation, the more representative is of equation {5}. For example, a full finite element model build with GEOMECH will be more representative than using an analytical solution of homogeneous linear elasticity.

Knowing the limitation of MVR, theoretically, there are two ways to improve the quality of the proxy. The first way is to use a higher-order model but this comes with a price. This can be achieved by designing higher-order experiments using DoE in combination with higher-order polynomial approximation. At this stage, it is important to pause and ask ourselves if the effort worthwhile given all the uncertainties in the input parameters.

A second less robust but sound common sense is to quantify ε in equation {5} by providing an upper and lower bound of the proxy. Suppose y_i, y_i^* are true responses and predicted responses from MVR, respectively. The unbiased estimator of a set of sample can be expressed as

$$\sigma_e = \sqrt{\frac{\sum_i (y_i^* - y_i)^2}{n-1}} \quad \{8\}$$

where n is the number of samples. The factor $n-1$ is called the Bessel's correction factor. The smaller the value of σ_e , the better is the fit. With this, equation {5} can be rewritten as

$$y = \varsigma x \pm k \sigma_e \quad \{9\}$$

where k is a real number that gives an adequate bounds to the proxy. The recommended value for k is 3 for a normally distributed distribution, which represents 99.7% of the solution spectrum and is conveniently called the 3-Sigma Rule.

2.5. Reliability Assessment

2.5.1. Limit State

One of the key steps in estimation of failure using any probabilistic method is to decide on what constitute unsatisfactory performance or failure. Mathematically, this is achieved by defining a Limit State, $g(X)$, such that $g(X) > 0$ means satisfactory performance, $g(X) < 0$ means unsatisfactory performance and $g(X)=0$ means a breakeven. X here is a vector of state variables. As a general rule of thumb in formulating the limit state, the following form is recommended:

$$g(R, Q) = R - Q \quad \{10\}$$

where $R = R(X)$ is the resistance and $Q = Q(X)$ is the load. Equation {10} is sometimes referred to as the 'safety margin' in geotechnical literature [5]. Thus, a system is considered in failure when $Q \geq R$ or $g(R, Q) \leq 0$. There are various reasons why equation {10} is favored. Apart from simplicity, it facilitates a convenient way to implement the limit state equation. The positive variable always represents resistance while the negative variable represents load. Simple

it may seem, equation {10} took the central stage of geotechnical debate in the last two decades, on the constitution of R and Q [10].

In general, guidelines provided by the regulatory agency can be used in creative ways to formulate the limit state equation. Figure 6 shows the UK Coal Board classification of subsidence damage due to mining activities. For illustration purpose, if it is decided that 0.6 ft of subsidence is unacceptable, then one replaces $R=0.6$ ft, and concentrate on finding an appropriate expression for Q ; in this case the subsidence.

Change of Length of Structure	Class of Damage	Description of Typical Damage
Up to 0.1 ft	1. Very slight or negligible	Hair cracks in plaster. Perhaps isolated slight fracture in the building, not visible on outside.
0.1 ft-0.2 ft	2. Slight	Several slight fractures showing inside the building. Doors and windows may stick slightly. Repairs to decoration probably necessary.
0.2 ft-0.4 ft	3. Appreciable	Slight fracture showing on outside of building (or one main fracture). Doors and windows sticking; service pipes may fracture.
0.4 ft-0.6 ft	4. Severe	Service pipes disrupted. Open fractures requiring rebonding and allowing weater into the structure. Window and door frames distorted; floors sloping noticeably; walls leaning or bulging noticeably. Some loss of bearing in beams. If compressive damage, overlapping of roof joints and lifting of brickwork with open horizontal fractures.
More than 0.6 ft	5. Very severe	As above, but worse, and requiring partial or complete rebuilding. Roof and floor beams lose bearing and need shoring up. Windows broken with distortion. Severe slopes on floors. If compressive damage, severe buckling and bulging of the roof and walls.

Figure 6: UK National Coal Board classification of subsidence damage [12]

If all applied loads and its uncertainties are represented by the variable Q and the total resistance R , the space of state variables is a two-dimensional space as shown in Figure 7. Within this space, one can separate the 'safe domain' from the 'failure domain'; the boundary between the two domains is described by the plane, $g(R, Q) = 0$.

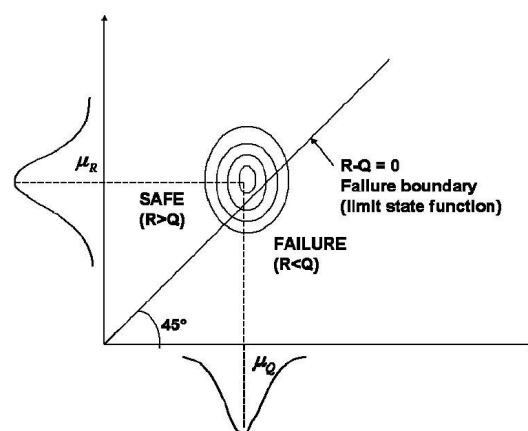


Figure 7: Plan view of safe domain and failure domain in a two-dimensional state space.

Theoretically, the probability of failure can be calculated by integration of the joint probabilistic density function over the failure domain (i.e. the region in which $g(R, Q) \leq 0$). However, integration of this failure probability is very difficult unless it is evaluated by means of brute force. Another important observation is that though the limit states $R - Q \leq 0$ and $R/Q \leq 1$ are mechanically equivalent, the integration leads to different values of probability of failure – a dilemma in the early development of the concept of reliability. As a consequence, an invariant reliability index is introduced by Hasofer-Lind [6] to circumvent this difficulty by giving it a new geometric meaning to the quantification of the integrity of structures.

2.5.2. First-Order Reliability Method

Instead of working entirely from the probabilistic viewpoint, Hasofer-Lind decided to tackle the problem from the geometric perspective as shown in Figure 8.

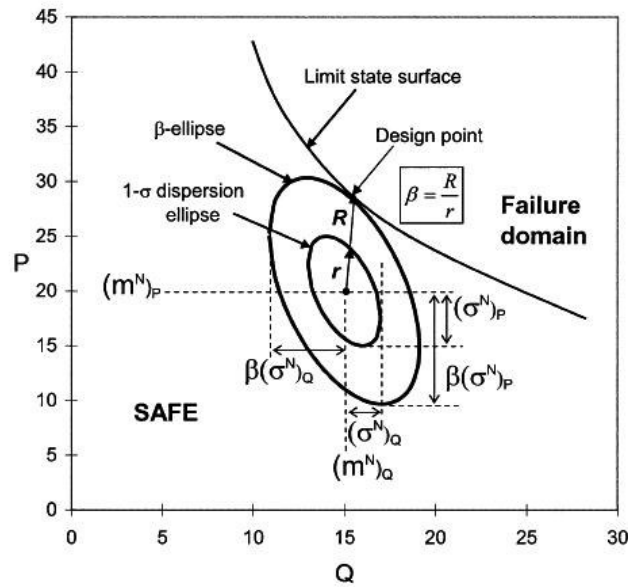


Figure 8: Design point and equivalent normal dispersion ellipsoids illustrated in the plane

Their 'correction' to the evaluation of the limit state is a point called the '*design point*'. The design point is the closest point drawn on the mean value of the joint probabilistic function to the failure surface $g = 0$. According to Figure 8, the reliability can be defined as the minimum distance in units of directional standard deviations from the mean-value point of the state variables to the boundary of the limit state surface [7] ,

$$\beta = \frac{R}{r} . \quad \{11\}$$

The design point is not known a priori and an iteration technique must be used to solve for the reliability index. Once the 'design point' is found and the reliability index computed, the failure probability can be calculated from the cumulative distribution function, Φ , of the standard normal variable according to [8] [9] ref. Fig. 4.

$$P_f = 1 - \Phi(\beta) = \Phi(-\beta) \quad \{12\}$$

The Hasofer-Lind reliability index transforms the original probabilistic problem into a geometry problem of constrained convex optimisation of the form

$$\text{Minimise } \beta = \sqrt{\left[\frac{x_i - \mu_i}{\sigma_i} \right]^T [\mathbf{R}]^{-1} \left[\frac{x_i - \mu_i}{\sigma_i} \right]}, \text{ subjected to the constraint } g(x) = 0 \quad \{13\}$$

in which $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_n)$ is the vector of mean values, μ_i, σ_i are the mean value and standard deviations of X_i respectively, $\mathbf{R}$ is the correlation matrix of $\mathbf{X}$. A more detailed description of the First-Order Reliability Method and the relevant literature on other reliability methods is discussed in the Appendix.

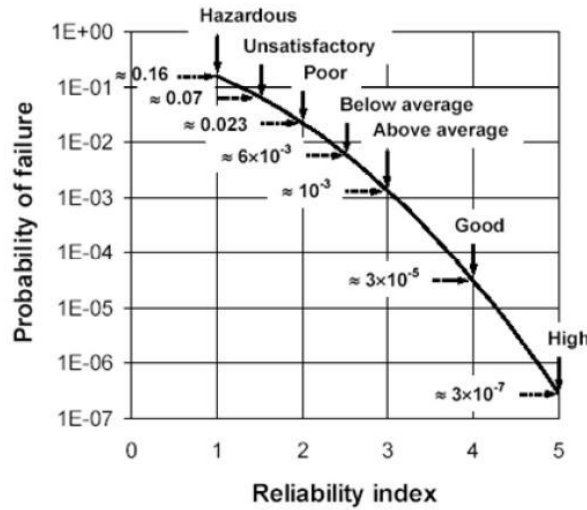


Figure 9: Relationship between reliability index and probability of failure
(adapted from US Army Corps of Engineers, 1997)

In order to appreciate the concept of reliability and its association with probability of failure, Figure 9 shows reliability index versus probability of failure chart issued by US Army Corps of Engineers for building, and their classification. For geotechnical design, it is difficult, if possible at all, to achieve a reliability index higher than 2 due to stochastic nature of geomaterial.

2.5.3. Simple Monte Carlo

No probabilistic procedure can be considered complete without involving Monte Carlo at some stage. Monte Carlo is a class of computational algorithms that rely on repeated random sampling to compute their results [11, Chap 7]. It is particularly suits for systems where there are uncertainties involve. Within the scope of the current report, only a very simple Monte Carlo algorithm with normal deviates is required. It is used to compute the probability of failure defined as

$$P_f = \frac{\text{Number of failed samples}}{\text{Total number of samples}} \quad \{14\}$$

Equation (14) is applied repeatedly for different values of control parameters, e.g. operating pressure to build the fragility curve. The reliability index can then be inverted according to Equation (12).

2.6. Fast Probabilistic Geomechanics Workflow

Sections 2.1~2.5 layout the required background information for probabilistic geomechanics. Figure 10 shows how each of these aforementioned components can be integrated to form a chain of analysis. The workflow consists of five major components or steps, namely (A) Geomechanics Requirements; (B) Experimental Design; (C) Geomechanics Engine, (D) Multi-Variate Regression and (E) Reliability Assessment. Each of these components should be viewed as independent but separately relevant mechanism that hinges the whole process together. In Figure 10, the solid line represents the flow of data between components and the broken line represents indirect relationships with the components. The function of each component will now be described.

2.6.1. Step A - Geomechanics Requirement

Geomechanics problems can be complicated due to inherent inhomogeneity and stochastic nature of geomaterials. Before any analysis can be carried out, it is necessary to plan for the **SMART** geomechanics requirements. In other words, the requirements need to be **S**pecific, **M**easurable, **A**ttainable, **R**ealisable and **T**imely.

Once the requirements are understood, one can proceed to gather the necessary data from field. The field data here is categorized into five major categories, (i) geologic topology, (ii) in-situ stress measurements, (iii) formation properties, (iv) reservoir model(s). This information is assumed available from exploration, analogue fields and open literature.

Subsequent to the collection of field data, a base geomechanical model can be built. The end product of Step A is thus a base geomechanical model.

2.6.2. Step B – Experimental Design

In the process of gathering information in Step A, one will come across ambiguous data and confronted with unavailability of information. This information forms the basis of the state variables that will go into the Experimental Design. At this stage, it is important to decide what are the likely state variables that influences the experimental outcome. The end product of Step B is a series of case studies to be run using the geomechanics model.

2.6.3. Step C – Geomechanics Engine

Once the design matrix is available, numerical models with the appropriate variation in the parameters can be prototyped accordingly and executed in batch-mode using available geomechanics engine, e.g. GEOMECH and QuickBlock. The geomechanics engine will be treated as 'black-box' in this report. The output from Step C is the simulation results or the post-processed responses.

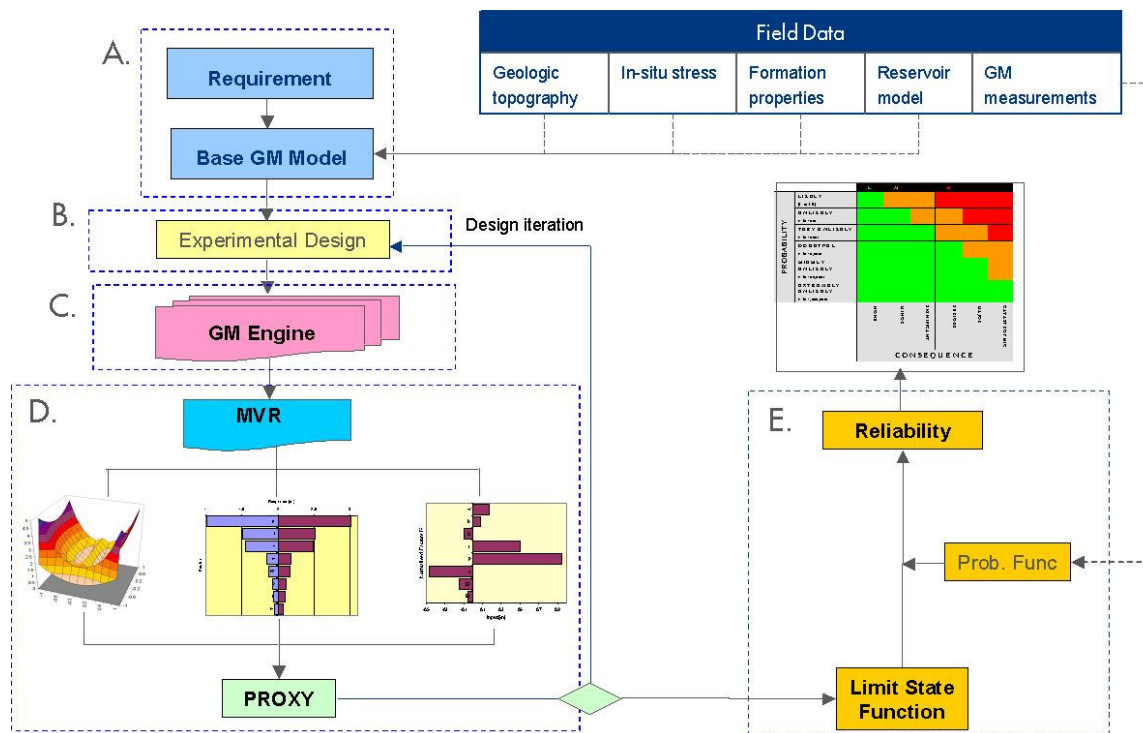


Figure 10: General workflow of fast probabilistic geomechanics

2.6.4. Step D - Multi-Variate Regression

From Step C, collections of post-processed responses are gathered. These responses can be subjected to the multi-variate regressional analysis in Step D to produce (a) state variables sensitivity, (b) state variables relative influence, and (c) response surfaces. A proxy of specific response can be built based on the collective information in (a), (b) and (c). A proxy is a linear approximation of specific observation in terms of the input state variables. It is the end product from Step D. In some situation, for very challenging geomechanical problem, it may be necessary to refine the proxy by going through Step B, C and D several times. This situation can arise when parameters that are initially thought to be sensitive to the responses are not and vice versa, or the engineer feels that there is need to increase/decrease the number of state variables.

2.6.5. Step E - Reliability Assessment

Once the proxy from Step D is obtained, a limit state equation can be defined accordingly. There is no specific rule on the formulation of the limit state equation. In general, it requires the engineer to condense his/her experience and working knowledge of the problem into an algebraic equation of operational significance. The rule of thumb in formulating limit state is given by equation (10).

To proceed, the Probabilistic Density Function (PDF) of each state variable need to be supplied. For normal/lognormal PDF, the mean, μ_i , and standard deviation, σ_i , of the of state variable x_i are sufficient. For non-normal probabilistic distribution, additional statistical parameters must be supplied in order to define the PDF. In this report, a standard normal distribution is used because it has a very simple one-to-one transformation in the form

$$Z = \frac{x - \mu}{\sigma} \quad \{15\}$$

where Z is the transformed value in the standard normal distribution curve with zero mean and unit standard deviation, see Figure 11. With this information, the limit state function can be solved e.g. by using Excel built-in solver.

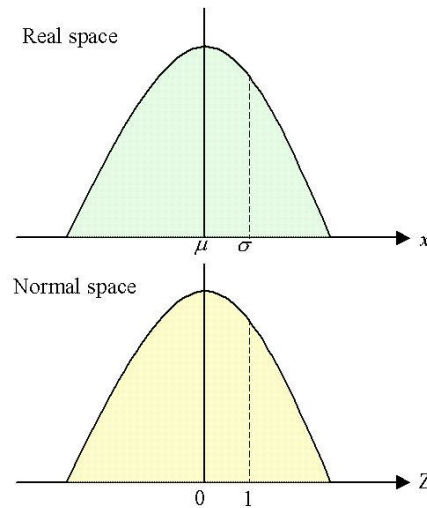


Figure 11: Transformation between real variable and normal variable

The output from Step E is the Design Point or the most probable point of failure. The Design Point indicates the values of the state parameters with the largest probability of failure. Once the Design Point is found, the fragility (failure) curve for different operating conditions can be generated using simple Monte Carlo.

2.7. Software and Resource Selection

The Fast Probabilistic Geomechanics Workflow consists of 5 independent but separately relevant steps. Each of this component constitutes a discipline of its own right. In order to link these individual steps to form a one-way communicable chain, a platform to integrate them into a system is necessary. Appendix 3 gives an assessment of the software and their pros and cons to put the Fast Probabilistic Geomechanics Workflow together. It was decided that at the current stage, Excel should be used as an integration platform to prototype the system and to gain insight through learning by doing approach. The DoE and MVR functionalities are provided by free GNU software from Ref. [18], while the Geomechanical Engine, FORM algorithm are built from scratch using Excel.

3. Feasibility Study of Fast Probabilistic Method

This section intended to illustrate, in a practical way, the steps involves in probabilistic geomechanics. It also intends to demonstrate how the idea developed in previous chapter is turned into tangible results.

To set the stage, a simple geomechanical model will be defined together with uncertainties in its input parameters. The DoE is used to spit out the design matrix with necessary combination of numerical simulations with varied input parameters. This design matrix is used to build the input datafiles necessary for execution by the Geomechanics Engine in batch-mode. The raw numerical solutions from the Geomechanics Engine is post-processed for re-inputing into the design matrix as responses. The completed design matrix is subsequently pipelined to a MVR solver to extract the proxy function. The upper and lower bound of the proxy is estimated a-priori within the MVR solver. The FORM algorithm took over the proxy to compute the design points. This is followed by Monte Carlo procedure to build the fragility curve which is superimposed on the design points to form a design chart.

3.1. Geomechanics model

The geomechanics model, depicted in Figure 12, is an axisymmetric model with a disk- shaped reservoir. The reservoir is buried at a depth D from the top surface, and extends at a radius R from its centre. The thickness of the reservoir is h . The base case of the geomechanical model is taken from Ref. [13], page 409. The uncertainty model can be characterised by 7 factors of uncertainties, as shown in Table 1. The low and high values of the uncertainties are assumed P_{10} and P_{90} of a normally distributed curve, and hence the mean and standard deviation of the state variables can be inverted accordingly.

In Table 1, shear modulus, Poisson's ratio and uniaxial compressibility is purposely chosen as state variables, knowingly that one of the three state variables is redundant. This is intentional to test the feasibility of the screening unit to sieve out the redundant variable. In this case, the redundant variable is shear modulus.

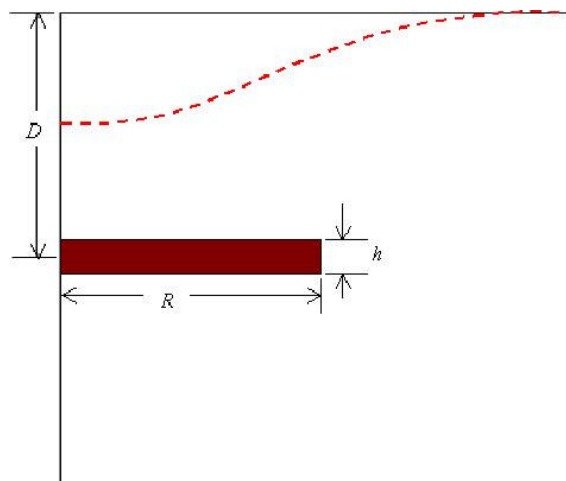


Figure 12: Geomechanical model for probabilistic analysis

The dotted line in Figure 12 is the expected surface subsidence profile due to pressure depletion. For simplicity sake, the maximum surface subsidence, which located at the bottom of the bowl is investigated. The surface subsidence of this model is given by Geertsma's analytical solution [14]

$$S = -2C_m(1-\nu)h\Delta p I_3(q) \quad \{16\}$$

where $I_3(q)$ is a complex function of elliptic integrals, S is the maximum subsidence and the other parameters are defined in Table 1.

Table 1: Upper and lower range of uncertainties for toy reservoir model

Parameters	Sym	P10	P90	μ	σ
Depletion Pressure (MPa)	p	5	15	10	3.9063
Shear Modulus (GPa)	G	1.5	2.5	2.0	0.1953
Poisson Ratio (-)	ν	0.2	0.3	0.25	0.0391
Uniaxial Compressibility (10^{-10} Pa^{-1})	C_m	1.2	2.2	1.7	0.39063
Reservoir Thickness (m)	h	80	120	100	15.625
Reservoir Radius (km)	R	1.8	2.2	2.0	0.1563
Reservoir Depth (km)	D	1.8	2.2	2.0	0.1563

3.2. DoE, Geomechanics Engine and MVR

Once the geomechanical model and the response parameters is chosen, the next step involves the generation of cases for experimentation. In this case, a Plackett-Burman design with 2 centre-points was used, resulting in 14 randomised cases, shown in Figure 13.

Exp #	$P(\text{MPa})$	$G(\text{GPa})$	ν	C_m	$h(\text{m})$	$R(\text{km})$	$D(\text{km})$	Resp_1
1	10.00	2.00	0.25	1.70	100.00	2.00	2.00	-0.164844
2	5.00	1.50	0.20	2.20	120.00	2.20	1.80	-0.14433
3	5.00	1.50	0.20	1.20	80.00	1.80	1.80	-0.049647
4	5.00	2.50	0.20	1.20	80.00	2.20	2.20	-0.049647
5	5.00	2.50	0.30	1.20	120.00	1.80	1.80	-0.065162
6	15.00	2.50	0.20	2.20	120.00	1.80	2.20	-0.38841
7	15.00	1.50	0.20	1.20	120.00	2.20	2.20	-0.223412
8	5.00	2.50	0.30	2.20	80.00	2.20	2.20	-0.079642
9	5.00	1.50	0.30	2.20	120.00	1.80	2.20	-0.113286
10	15.00	2.50	0.30	1.20	120.00	2.20	1.80	-0.206654
11	10.00	2.00	0.25	1.70	100.00	2.00	2.00	-0.164844
12	15.00	1.50	0.30	1.20	80.00	1.80	2.20	-0.123585
13	15.00	2.50	0.20	2.20	80.00	1.80	1.80	-0.273059
14	15.00	1.50	0.30	2.20	80.00	2.20	1.80	-0.252578

Figure 13: Excel output of Plackett-Burman design

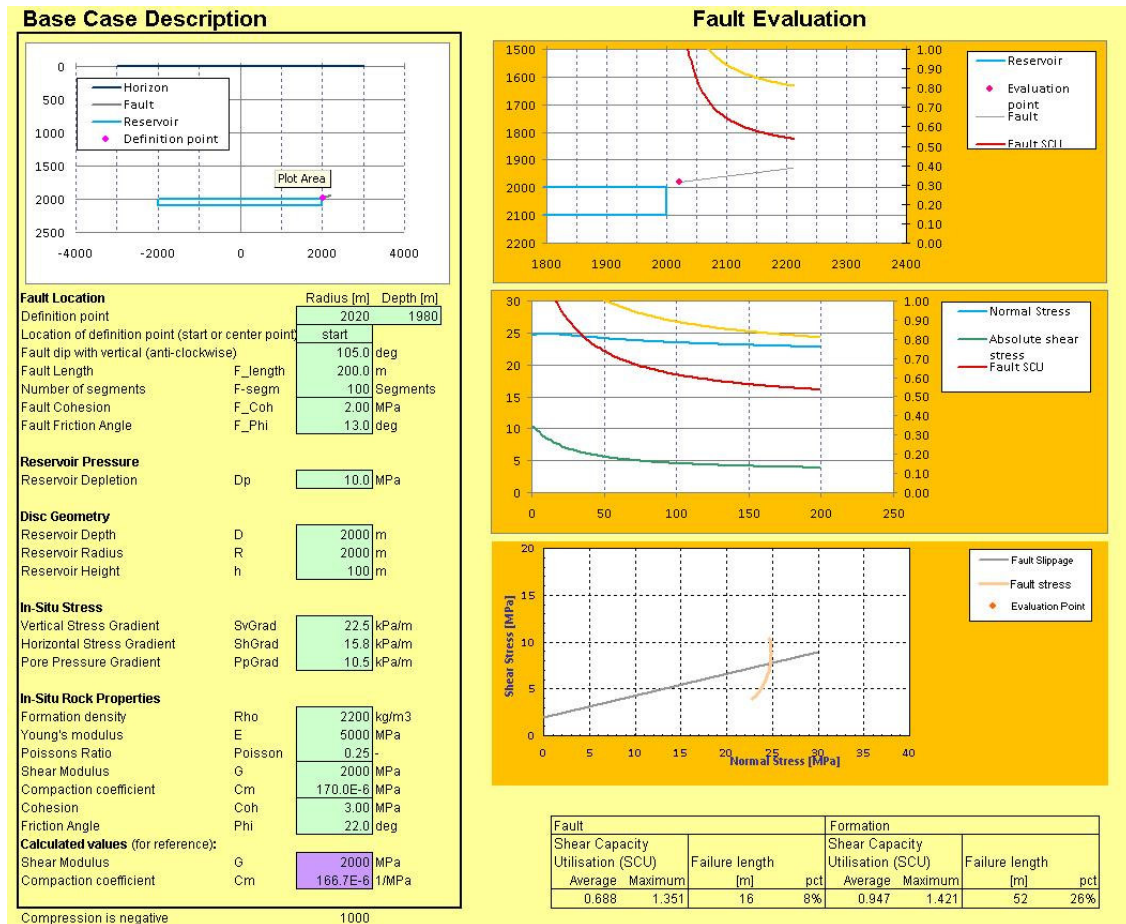


Figure 14: Excel based Geertsma geomechanical engine

The geomechanical engine used for the exercise is the Excel-based Geertsma model, as shown in Figure 14. Detailed information about Geertsma's model can be found in Ref. [13,14,25].

The surface displacement obtained from the geomechanical engine is depicted in column 9, under the heading Resp_1 in Figure 13.

Once the solution is obtained, MVR can be carried out. This results in the following proxy function for the subsidence in metre, S , (truncated here to 2 significant digits):

$$S = 0.15 - 0.016p - 0.026G + 0.48v - 0.89C_m - 0.0013h + 0.024R + 0.0056D \quad \{17\}$$

From equation (17), a tornado chart and a parameters relative influence chart can be plotted as shown in Figure 15 and Figure 16, respectively. The tornado chart characterises the sensitivity of each factors with respect to the response. Notice that the mean-value of the response in Figure 15 is approximately 0.15 m, which is given by the first term in the RHS of equation (17). The parameters relative influence chart is shown in Figure 16. It quantitatively characterises the influence of each factors' uncertainty with respect to the response. A direct consequence of this is that the wider the spectrum of uncertainty, the higher is the relative influence.

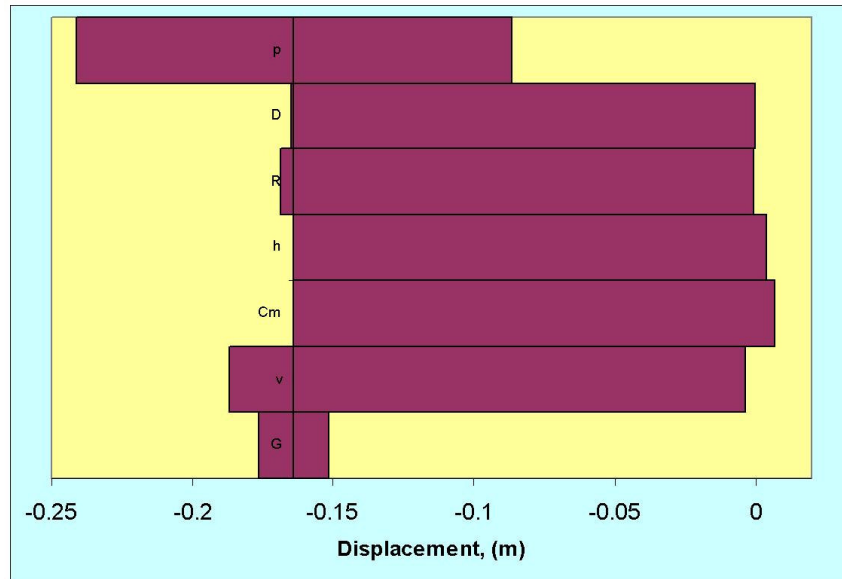


Figure 15: Tornado chart characterising sensitivities of each factor

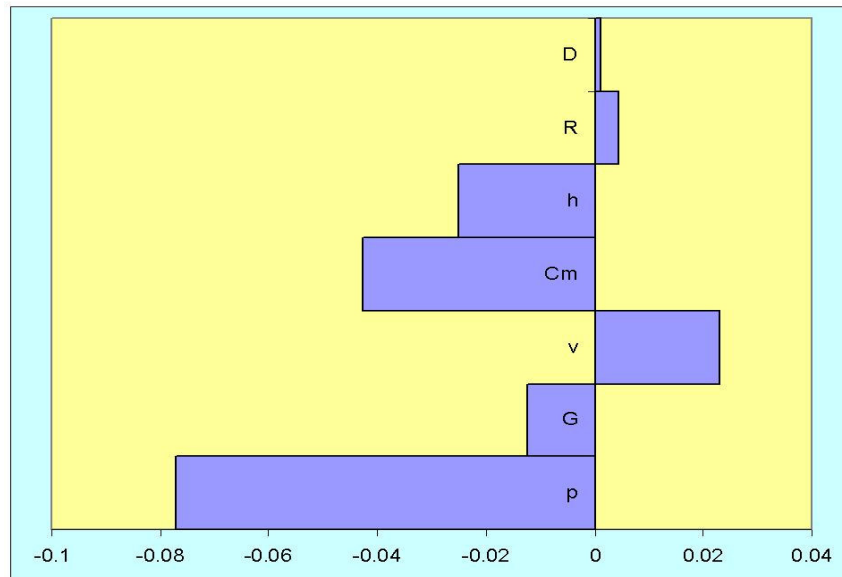


Figure 16: Relative influence chart characterising the relative impact of each factors in the x-axis

By careful study of Figure 15 and Figure 16, it is not surprising to conclude that the state variable G may be a redundant parameter because in both chart, its sensitivity and relative influence are both minimal. This is verified by the analytical solution in equation {16}. The relatively small influence of parameter D and R in Figure 16 needs to be explained. Notice that the solution is very sensitive to the change in D and R in Figure 15 while its influence is minimal. This is because the range of uncertainty for both state variables is very small in relation to their mean values, a deviate of $\pm 10\%$, see Table 1.

In order to estimate the upper/lower bounds to the proxy function, the true solutions (equation (16)) can be plotted against the predicted solutions (equation (17)). This result is shown in Figure 17. If the match is perfect, the dotted data points will fall on the straight line. If the match is not perfect, the deviation represents the error of prediction. This error can be used as a priori-

estimate to establish the upper/lower bounds of the prediction using Eq. (8). Based on Figure 17, the unbiased estimator is computed to be $\sigma_E = 0.018103\text{m}$. Figure 18 depicted the true solutions against the predicted solutions using Monte Carlo simulation of 8000 samples. The upper/lower bounds established from the experimental design simulation is superimposed onto Figure 18 to check for its validity. As expected, apart from a few outliers, the majority of the solutions (theoretically 99.7%) fall within the established bounds.

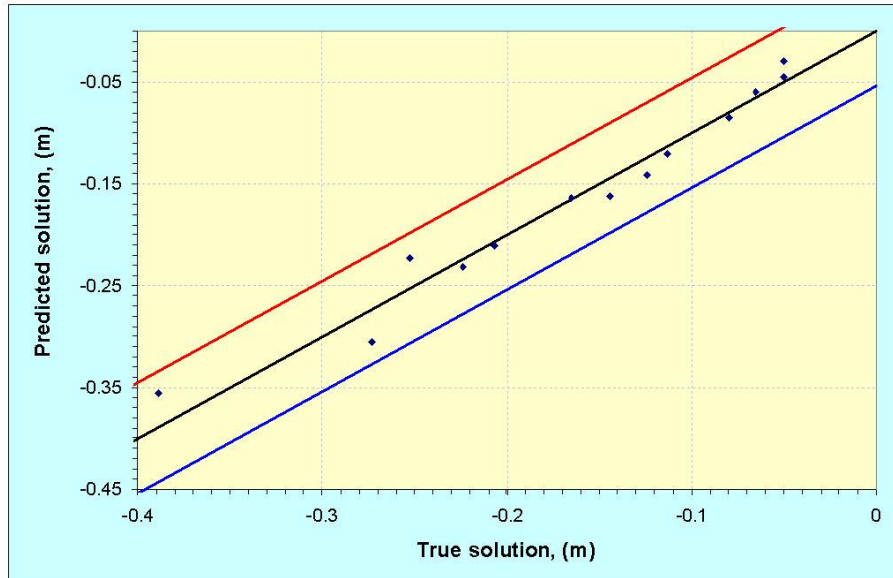


Figure 17: True solution against the predicted solution with upper/lower bounds

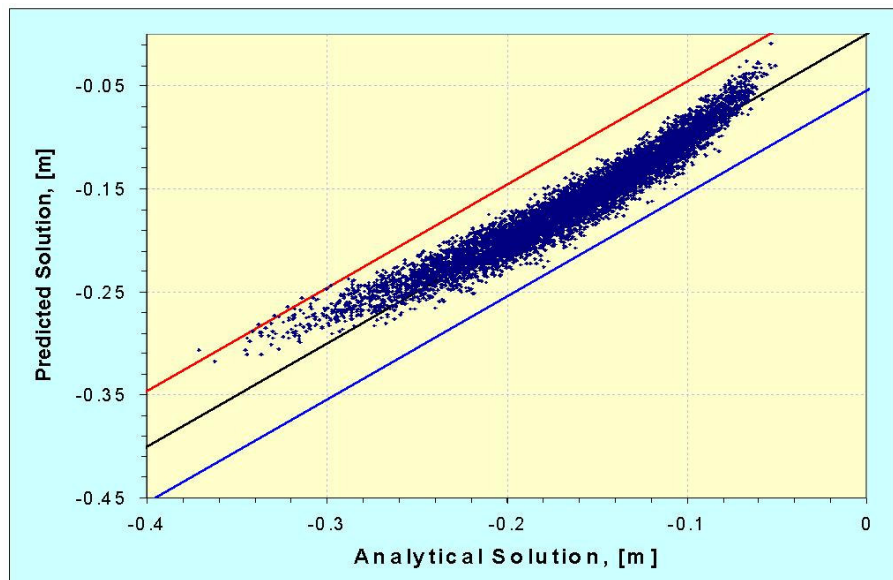


Figure 18: True solution against the predicted solution using Monte Carlo simulation. Upper/lower bounds established from experimental design simulation.

The unbiased estimator from Figure 18 is calculated to be $\sigma_E^{MC} = 0.013556$ m, which is smaller than σ_E . As the sample size increases, the error will approach a constant value. An additional point is that σ_E is a priori-estimate approaching the true error from above.

Figure 18 also reveals the nature of the surface displacement solution and its relation to the predicted proxy function solution. Even though the data points cluster around the line $y=x$, the cluster bends upwards in its trailing edges. This informs us that the surface displacement, according to Geertsma solution, is non-linear despite linear elasticity. This is not surprising since the general solutions for partial differential equations for displacement are usually expressed in sum of power series in terms of spatial variables.

3.3. Reliability Assessment

One of the key steps in estimating failure using any probabilistic method is to decide on what constitute unsatisfactory performance or failure. Mathematically, this is achieved by defining a Limit State according to equation (10). In the current exercise, the resistance is simply chosen to be 0.25 m. Figure 19 shows the FORM excel template for this calculation, and the reliability is computed to be $\beta = 1.11$ with a probability of failure $P_f = 13\%$. The design point for each state variables are automatically computed using the Excel Add-in Solver and listed under the column x_i^* .

No	Parameters	Sym	Distribution	Para1	Para2	Para3	Para4	x_i^*	n_i
1	Depletion pressure	p	Normal	10.00	3.91	0.00	0.00	13.5319	0.9042
2	Shear modulus	G	Normal	2.00	0.20	0.00	0.00	2.0142	0.0729
3	Poisson's ratio	v	Normal	0.25	0.04	0.00	0.00	0.2395	-0.2692
4	Uniaxial compaction	Cm	Normal	1.70	0.39	0.00	0.00	1.8950	0.4991
5	Reservoir thickness	h	Normal	100.00	15.63	0.00	0.00	104.5790	0.2931
6	Reservoir Radius	R	Normal	2.00	0.16	0.00	0.00	1.9917	-0.0532
7	Reservoir Depth	D	Normal	2.00	0.16	0.00	0.00	1.9980	-0.0126
8			Normal						
9			Normal						
10			Normal						

LIMIT STATE	$g(\bar{x})$	-6.5503E-15
RELIABILITY	β	1.1105
FAILURE PROBABILITY	P_f	0.1334

Figure 19: FORM excel template. Para1 and Para2 are mean and standard deviation while Para3 and Para4 are not used currently. x_i^* and n_i are computed failure points in the real space and normal space, respectively.

Figure 20 shows the reliability index versus depleting pressure for upper bound, mean and lower bound cases, superimposing on the limit state depletion pressure (triangle data point). The reliability chart is computed by applying Monte Carlo algorithm to the proxy function. To ensure that the Monte Carlo procedure generates random numbers that conforms to the specific probabilistic density function according to the input in Table 1, a Reject algorithm is used [11].

The Reject method is augmented with a cut-off to ensure that numbers beyond the upper/lower limits are not used in the computation. The code is implemented in Excel Visual Basic.

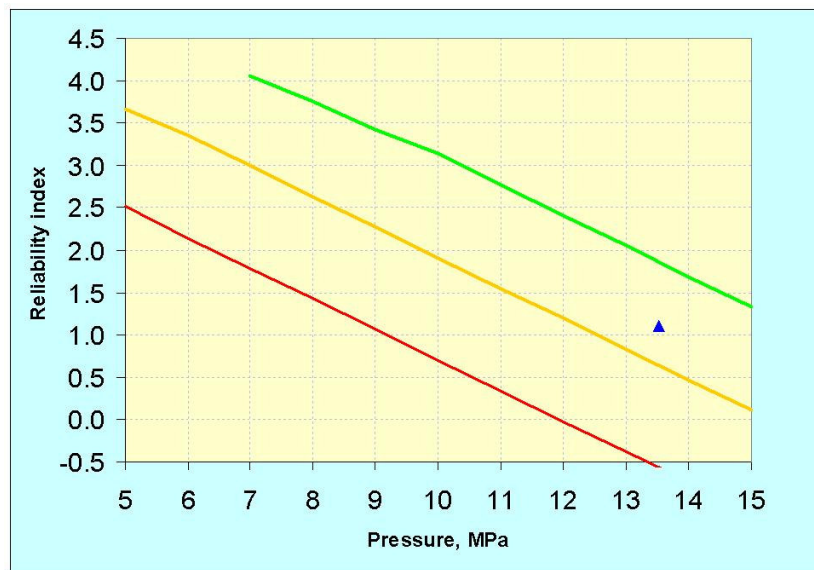


Figure 20: Reliability lines for high, mean and low cases of the state parameters using Monte Carlo, superimposing on the limit state depletion pressure (triangle)

It is necessary at this point to explain why the limit state (triangle data point) computed using FORM does not fall 'right on' the mean reliability curve. The reason is because of the cut-off algorithm used in the Monte Carlo procedure to avoid unphysical numbers. To confirm this idea, a validation test was carried out using Monte Carlo procedure with different sampling sizes, without imposing the cut-off. The results were compared with FORM in Figure 21. The comparison showed that a minimum of 100,000 and above sampling is required before a stable solution is obtained that matches the FORM reliability index to two significant figures.

A few important reasons behind this use of mixture of FORM and Monte Carlo now needs to be explained, and most of them are due to practicality and resources available. The main reason for using FORM is to obtain the design points. The design point informs the locality of the failure parameters via optimisation of the limit state. This optimisation function cannot be delivered by Monte Carlo.

On the other hand, even though FORM can compute the reliability index at a speed magnitude of orders faster than Monte Carlo, its nature is not a random method as per se but a constrained convex optimisation problem. When too many constraints are imposed (e.g. requiring it to test different pressure values and making sure that all the state variables stay within acceptable range) the mathematical problem becomes over-prescribed and the algorithm fails in Excel. Hence, at the current stage, FORM is only suitable for finding the limit states (design point) given that all the state variables are random.

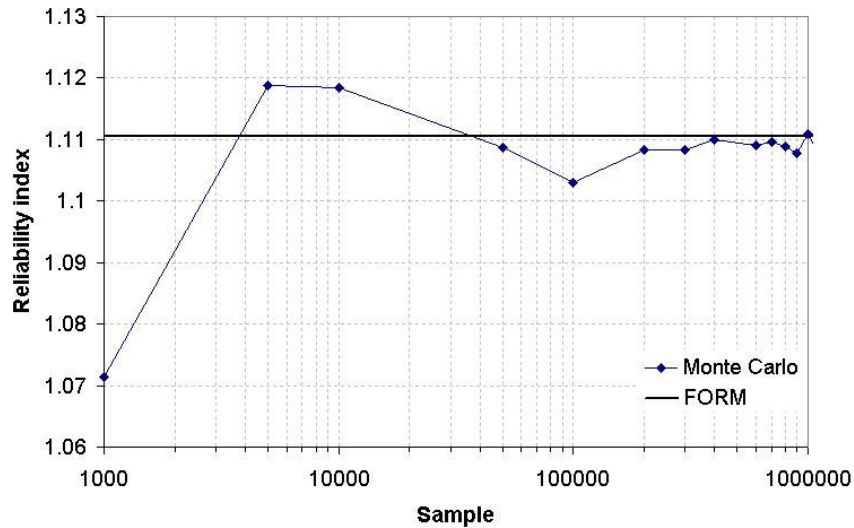


Figure 21: Reliability index using Monte Carlo versus sampling size requirement

Figure 22 shows the probability of failure versus depleting pressure for high, mean and most likely cases.

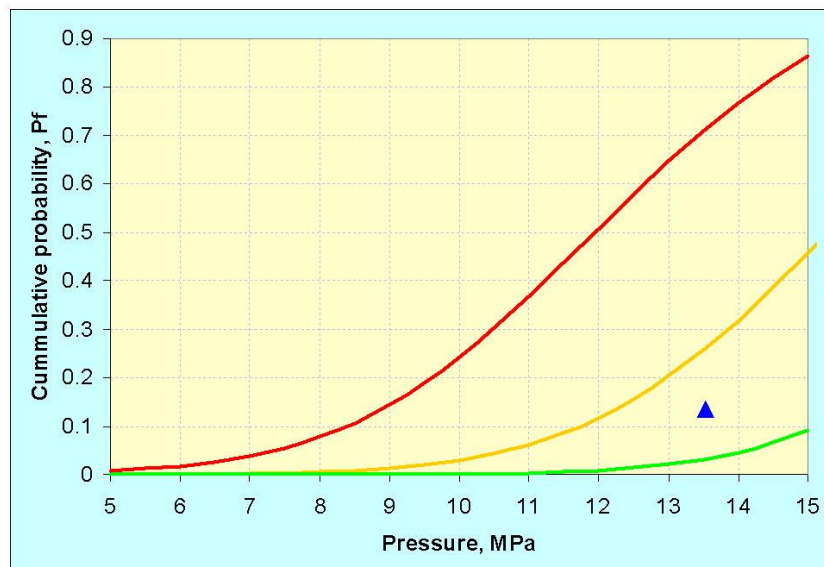


Figure 22: Depletion pressure fragility curve for high, mean and most likely cases, superimposing on the limit state depletion pressure (triangle)

With the help of Figure 20, a design chart can be proposed as shown in Figure 23, where the range of depletion pressure is compartmented into a safe, cautious and unsafe zone. The upper limit of the safe zone is taken as when $\beta = 1$, and the lower limit of the unsafe zone is computed from FORM.

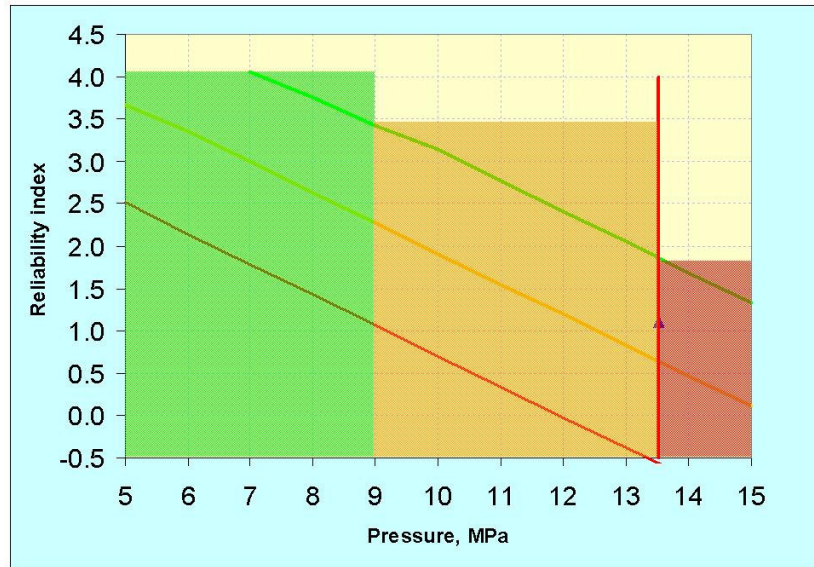


Figure 23: Proposed design chart for safe, caution and unsafe depletion pressure

3.4. Concluding Remarks

In this chapter, a feasibility study of the fast probabilistic geomechanics has been demonstrated by transforming the idea into a tangible example. The geomechanical model has been carefully chosen to ensure that the model problem is not overwhelming the main objective of the probabilistic analysis. A total of seven geomechanical state variables are studied as uncertainty in the model and the maximum reservoir subsidence is chosen as state response. With the utility of Design of Experiment and Multi-Variate Regression, a proxy for the subsidence model is built. By formulating the proxy in a creative way, limit state behaviour for maximum allowable subsidence can be investigated. Using First-Order Reliability Method, the Design Point for maximum allowable subsidence is found and its fragility curves constructed using Monte Carlo method.

4. General Conclusions and Future Research

The properties of geomaterials vary within a considerable range. If the mean or characteristic values of the parameters are used in deterministic analysis, the underlying uncertainties of the parametric values cannot be explicitly reflected in the final response. As the geomechanical model becomes increasing complex, the number of uncertainties increases accordingly. Consequently, the complexities of the interplay between the factors and the responses increases in an exponential fashion. At a tipping junction, it becomes difficult for one to distinguish and rank in a quantitative manner the combinatoric effects of each uncertainties that contribute to a specific response.

In order to predict a stochastic system responses, two serials tactics were proposed in this report to deal with; firstly the complexities of the inter-woven relationships between the state variables and its responses, and secondly probabilistic approach to capture the parameters variability. The first tactic, named the screening unit, aims to unravel the complexities of the input parameters and its responses. In other words, it seeks to sieve out the important parameters from the potentially important parameters. The screening is achieved via the Design of Experiments and Multi-Variate Regression. Once the thread of these complexities has been disentangled, probabilistic method took over. Using a combination of First-Order Reliability Method and Monte Carlo, the design point of the stochastic system is located and the corresponding fragility curve drawn for reliability assessment.

In summary, this report has achieved the objectives and scopes outlined for this project and are recaptured herein as follows:

1. A brief overview of Design of Experiment (DoE) and the Multi-Variate Regression (MVR) to provide background information needed as the stepping-stone for the geomechanics screening unit.
2. A systematic way to extract the 'proxy' or the approximating functions representative of the uncertainties in the geomechanics model. As a first step in this project, it will be limited to first-order approximation.
3. Using geomechanics tool to investigate the response of a specified type of failure mechanism, in this case, surface subsidence due to production. The reasons for choosing this mechanism are because of its simplicity in identifying the failure and availability of analytical solution.
4. Institute a set of workable failure criteria, ranging from the very conservative to a very 'progressive' one. This is effect to investigate the implication of failure criterion on the reliability index.
5. Establish a systematic way to quantify the impact of each uncertainties and a ranking system for the sensitivity of each.
6. Estimating the upper and lower error bound of the approximating function.
7. Perform safety reliability assessment based on the approximating function and identifying the critical states.

4.1. Learning

It is important to summarise the learning from this project such that the lessons learnt can be used as stepping stone for the RELIACON project and DRMS/SRMS.

4.1.1. *Limit state*

The reliability index is directly dependent on the formulation of the limit state. Since there is only rule of thumb for its formulation as a guideline, different limit state formulation should be carried out to investigate its impact on the reliability. As noted in Ref.[10], the definition of Load and Resistant is very subjective and may vary for different application.

4.1.2. *Over-constraints in FORM*

FORM is used to compute the Design Point, which is subsequently fed as a guideline for operational design. Due to the way FORM is currently implemented, it sometimes gives unphysical Design Point, especially when using normal probabilistic function. In order for the Excel Solver to zoom-in the limit state, it sometimes have to search the solution from the entire probabilistic continuum extending from -ve infinity to +ve infinity. One way to circumvent this is to use the lognormal probabilistic function that prevent the Solver to search in the -ve continuum but this only solves half of the problem. On the other hand, constraining the search may hamper the Solver ability by over-prescribing it. Thus, it should not be surprising that FORM may fail in some situation.

4.1.3. *Software translator*

One major hurdle that repeatedly arises in this project is the integration of various components that hinges the workflow together. This is most serious in the pre- and post-processing of the geomechanics engine. The pre-processing here refers to the reading of the design matrix and the preparation of the geomechanics input data files in batch-mode. The post-processing here refers to the further processing of the raw output from geomechanics engine, encapsulating them into a form that is usable for constructing the proxy. Another identified critical path is the communication from MVR to the Reliability Assessment.

4.2. **Future Research**

4.2.1. *Limit state formulation for containment*

For reservoir containment, either in the production of hydrocarbon or storage of carbon/sour gas, the following mechanisms that could potentially leads to loss of containment due to geomechanics has been preliminarily identified: (a) reactivation of faults; (b) loss of cap rock integrity; (c) fracturing due to injection and (d) loss of borehole integrity. However, these generic identification needs to be armed with sufficient technical and operational 'know-how' if any realistic progress is to be made. Key questions such as favourable spectrum of cap rock integrity, the constitution of fault failure etc needs to be researched and precise definition attached. On the other hand, a too conservative criteria tends to shut-away opportunity while a too 'progress' limit state could lead to disastrous consequences. Thus a balanced and strictly scientific point is necessary in formulating the limit state.

4.2.2. *Software integration*

Four major critical paths in the prototyping of the fast probabilistic geomechanics workflow has been identified, namely (a) integration between the Experimental Design matrix and its automatic batch-mode input to geomechanics engine; (b) integration between the geomechanics engine output and the required post-processing as input responses to the Design matrix; (c) automation of batch-mode processing of multiple proxy approximation in the Multi-Variate Regression and (d) integration between the Multi-Variate Regression and Reliability Assessment for multiple limit state investigation.

4.2.3. *Directional Adaptive Response Surface*

The current approach using the Experimental Design can be augmented by using Directional Adaptive Response Surface (DARS) [15]. Rather than manually creating the Experimental Design matrix and run the simulation model sequentially, DARS uses specially formulated algorithms to automatically search for the most appropriate simulation. During the iterative search, it builds the response surfaces and limit state parallelly. This is particularly suitable for numerically intensive problems where a large number of state variables are involved.

4.2.4. *Random finite element with important sampling*

According to EuroCode 7 [16], for geotechnically challenged category 3 structures (e.g. underground storage of carbon), finite element Monte Carlo modelling is an essential requirement. Some researchers called it Random Finite Element Method. The reason it is emphasized because it is well known that the true limit state of a geotechnical system is non-linear. It is hoped that, with Monte Carlo Method, the inherited non-linearity can be revealed. As noted in the literature survey (see Appendix), combining finite element modelling and Monte Carlo Method is a very expensive venture, both timely and financially. Hence, for long-term research, a clever way must be devised to optimise the Monte Carlo sampling techniques so as to minimize the number of simulations. One of the key directions that could be pursued is Important Sampling. It is a technique for estimating properties of a particular distribution, while only having samples generated from a different distribution rather than the distribution of interest [17]. In other words, rather than samples from the whole spectrum of probabilistic distribution, samples only in the area where it is needed. Since the limit state is obtainable from FORM, it is natural to sample the uncertainties in the region near the vicinity of the limit states.

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Appendix 1. Design of Experiments and Response Surface

A1.1. Design of Experiments

Design of experiments is the design of an *information-gathering-screening* exercise where variation is present, whether under the full control of the experimenter or not. Often the experimenter is interested in the response of some process or of a particular system. It is thus a discipline that has very broad application across all the natural and social sciences. In order to adopt the DoE concept under the current framework of geomechanics and confine the scope of the research within the present context, it is important to restrict the present design factors to continuous quantitative variables because they are measurable and can be described easily by real numbers. Qualitative factors will not be considered.

Paramount to the goal of DoE in geomechanics is the designs used for screening. They are normally done to gain insight into which factors are important in a particular safety issue, which could be caprock integrity, surface subsidence or seal leakage, given a depletion/injection scenario. Consequently, screening allows one to narrow down the long list of so-called 'potentially important' factors into those that are *really* important with a known amount of statistical confidence. In short, DoE for screening are used to determine if a factor is important or not given all the uncertainties associated with the input.

The procedure by which one would intuitively accomplish a screening design is by running a given factor at two levels (high-low levels) and observe the variation of this factor's impact on the response. The simplest design for accomplishing this called the two-level full factorial design, in which a brute force approach is taken and every factor is run with all the other factors at all their possible settings. The immediate short coming of this design is that the number of experiments required is exponentially proportional to the number of factors according to 2^n , where n is the number of factors considered. For example, $2^6=64$ runs is for a two-level, full factorial design with six factors, and one can thus quickly run up to a very large resource requirement for runs with only a modest number of factors.

A1.1.1. Fractional Factorial Design

The ASQC (1983) Glossary & Tables for Statistical Quality Control defines fractional factorial design in the following way: "*A factorial experiment in which only an adequately chosen fraction of the treatment combinations required for the complete factorial experiment is selected to be run.*" The basic purpose of a fractional factorial design is to economically investigate cause-and-effect relationships of significance in a given experimental setting. This does not differ in essence from the purpose of any experimental design. However, because we are able to choose fractions of a factorial design, and hence be more economical, we also have to be aware that different factorial designs serve different purposes. Within the current context, the research is limited to designs of resolution III, which main objective is seek to screen out the important main effects from the many less important others. For this reason, these designs are often termed main effects designs, or screening designs in the statistical community. It is worth mentioning that designs of resolution five and higher are used for focusing on more than just main effects in an experimental situation. These designs allow one to estimate interaction effects and permits estimation of a full

second-order quadratic modelⁱ. There are currently powerful commercial software that allows higher-order resolution experiments to be designed, and thus will not be further elaborated here. To understand the fundamental of fractional factorial design, consider the two-level factorial design for three factors, namely the 2^3 design, represented by the cube in Fig. A1.1.

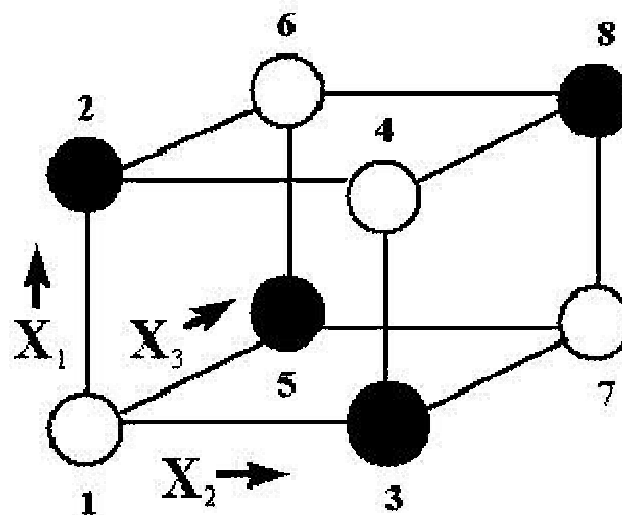


Figure A1.1: A 2^3 Full Factorial Design; Factors X_1 , X_2 , X_3 . (The arrows show the direction of increase of the factors. Numbers '1' through '8' at the corners of the design cube reference the 'Standard Order' of runs)

	X_1	X_2	X_3	Y
1	-1	-1	-1	Y_1
2	+1	-1	-1	Y_2
3	-1	+1	-1	Y_3
4	+1	+1	-1	Y_4
5	-1	-1	+1	Y_5
6	+1	-1	+1	Y_6
7	-1	+1	+1	Y_7
8	+1	+1	+1	Y_8

Table A1.1: A 2^3 two-level, full fractional design table showing runs in standard order, plus observations.

ⁱ If required, the second-order quadratic model can be used in conjunction with Second Order Reliability Method (SORM) described in First Progress Report.

The right-most column of the table lists y_1 through y_8 to indicate the responses measured for the experimental runs when listed in standard order. The numbers entered in the "y" column is used to illustrate the responses. From the entries in the table we are able to compute all 'effects' such as main effects, first-order 'interaction' effects, etc. For example, to compute the main effect estimate ' c_1 ' of factor X_1 , we compute the average response at all runs with X_1 at the 'high' setting, namely $(1/4)(y_2 + y_4 + y_6 + y_8)$, minus the average response of all runs with X_1 set at 'low,' namely $(1/4)(y_1 + y_3 + y_5 + y_7)$. That is,

$$c_1 = (1/4)(y_2 + y_4 + y_6 + y_8) - (1/4)(y_1 + y_3 + y_5 + y_7)$$

Suppose, however, that we only have enough resources to do four runs, it is still possible to estimate the main effect for X_1 and there are even different choices of the four runs that will accomplish this, and this idea forms the core fundamentals of fractional factorial design.

For example, suppose we select only the four light (unshaded) corners of the design cube. Using these four runs (1, 4, 6 and 7), we can still compute c_1 as follows:

$$c_1 = (1/2)(y_4 + y_6) - (1/2)(y_1 + y_7)$$

Similarly, we would compute c_2 , the effect due to X_2 , as

$$c_2 = (1/2)(y_4 + y_7) - (1/2)(y_1 + y_6) \text{ or}$$

Finally, the computation of c_3 for the effect due to X_3 would be

$$c_3 = (1/2)(y_6 + y_7) - (1/2)(y_1 + y_4)$$

We could also have used the four dark (shaded) corners of the design cube for our runs and obtained similar, but slightly different, estimates for the main effects. In either case, we would have used half the number of runs that the full factorial requires.

A1.1.2. Plackett-Burman Designs

In 1946, R.L. Plackett and J.P. Burman published their now famous paper "The Design of Optimal Multifactorial Experiments" in *Biometrika* (vol. 33). This paper described the construction of very economical designs with the run number a multiple of four (rather than a power of 2). Plackett-Burman designs are very efficient screening designs when only main effects are of interest.

Plackett-Burman (PB) designs are used for screening experiments because, in a PB design, main effects are, in general, heavily confounded with two-factor interactions. The PB design in 12 runs, for example, may be used for an experiment containing up to 11 factors.

PB designs also exist for 20-run, 24-run, and 28-run (and higher) designs. With a 20-run design you can run a screening experiment for up to 19 factors, up to 23 factors in a 24-run design, and up to 27 factors in a 28-run design. These Resolution III designs are known as *Saturated Main Effect* designs because all degrees of freedom are utilized to estimate main effects. The designs for 20 and 24 runs are shown below.

A1.2. Response Surface Methodology

Response Surface Methodology (RSM) is a method used to locate the optimum/sub-optimum value of the response and involves in the final stages of experimental design. The most extensive applications of RSM are in the particular situations where there is uncertainties in the input

variables potentially influencing some performance measure or quality characteristic of the process.

The field of response surface methodology consists of the experimental strategy for exploring the space of the process or uncertain variables to develop an appropriate approximating relationship called *proxy* between the yield and the process variables. In this report, parallel with the EPT-RHS Containment Analysis for geomechanics screening, it will focus on the statistical modeling to develop an appropriate approximating model between the response y and uncertain variables $\xi_1, \xi_2, \dots, \xi_n$, where n is the number of uncertainties. Note that the response y in this case represents a collection of observations at a fixed observation point, e.g. displacements, compaction, strains and stresses at coordinate (x_1, y_1, z_1) .

In general, the relationship between a response and its processes can be written as

$$y = f(\xi_1, \xi_2, \dots, \xi_n) + \text{err} \quad \{A1\}$$

where f is unknown and perhaps very complicated, and err is an error term that represents other sources of variability not accounted for. The error includes effects such as measurement error on the response, background noise, the effect of other variables and is treated as a statistical error.

The variables $\xi_1, \xi_2, \dots, \xi_n$ are called the natural variables, because they are expressed in the natural units of measurement, such as degrees Celsius, MPa etc. In RSM, it is more convenient to transform the natural variables to coded variables $\varsigma_1, \varsigma_2, \dots, \varsigma_n$, which are defined to be dimensionless with mean zero and the same standard deviation. In terms of the coded variables, the response function (A1) will be

$$\eta = g(\varsigma_1, \varsigma_2, \dots, \varsigma_n) + \text{err}(\varsigma) \quad \{A2\}$$

Because the form of the true response function η is unknown, it must be approximated. The polynomial used to approximate function η is the following second-order polynomial

$$\eta = \beta_0 + \sum_{i=1}^n \beta_i \varsigma_i + \sum_{i < j} \beta_{ij} \varsigma_i \varsigma_j + \sum_{i=1}^n \beta_{ii} \varsigma_i^2 + \text{err}(\varsigma_i^3) \quad \{A3\}$$

The β_0 is called the base value representing the base case or reference case, the term β_i is called the main effects in the literature of DoE representing the first-order effect of the processes; β_{ij} is the interaction between variables and β_{ii} is the secondary effects. Notice that the main effect terms are linear. Adding the interaction term into (A3) introduces slight curvature in η and with the second-order terms, the response function becomes curved even more.

At this point, it is worthwhile to take a step back and contemplate on the original usage of RSM. Traditionally, the RSM was designed to optimise manufacturing processes where ς_i are variables which could be tuned to yield the optimum η . In geomechanics, the reality is very different even though there is a shared relevance. In reality, the rock mechanical/thermal properties, the reservoir stratigraphy etc., are not susceptible to tuning. There are limited input parameters e.g. depletion pressure and temperature that could be controlled and managed artificially. From the geomechanics specialist standpoint, reservoir depletion pressure and temperature are *optimised* design by reservoir recovery specialist. Under this circumstances, pinpointing on the geomechanics input variables with second-order polynomial to a single value does not make much logical sense since one could not change the nature of the reservoir rock. It is within this context that it is only rational to simplify Eq. (A3) into

$$\eta = \beta_0 + \sum_{i=1}^n \beta_i \zeta_i \quad \{A4\}$$

where only the main impact of each input variables are investigated. In DoE literature, Eq. (A4) is also called the equation for screening.

Appendix 2. Literature Review of Probabilistic Geomechanics

The objective of this Appendix is to give an overview of the literature of the reliability method commonly used in the geomechanical analysis of the underground structures. Three methods will be reviewed, namely the (1) First-Order Second Moment Method, (2) The First-Order Reliability Method and (3) the Random Finite Element Method. Pros and cons of each methods will be discussed and its limitations analyzed.

A2.1. The First-Order, Second-Moment Approach (FOSM)

The First-Order, Second-Moment (FOSM) approach **Error! Reference source not found.** provides analytical approximations for the mean and standard deviation of a parameter of interest as a function of the mean and standard deviations of the various input factors, and their correlations.

Consider Y to be a function of random variables $X = \{X_1, X_2, \dots, X_n\}$; that is

$$Y = \{X_1, X_2, \dots, X_n\} \quad \{A5\}$$

In the general case, $\{X_1, X_2, \dots, X_n\}$ are correlated with covariance matrix $[C]$, i.e.

$$[C] = [\sigma][R][\sigma] \quad \{A6\}$$

where $[\sigma]$ is a diagonal matrix of standard deviations and $[R]$ is the (positive-definite and symmetric) correlation matrix with diagonal elements $R_{ii} = 1$ and non-diagonal $R_{ij} = \rho_{ij}$ (ρ_{ij} is the correlation coefficient between variables i and j).

In order to evaluate the mean and standard deviation of Y , the joint probability density function of $\{X_1, X_2, \dots, X_n\}$ is needed. However, in many practical applications the available information about the random variables is limited to their mean and variance. The approximate mean and variance of the function Y may still be estimated by a Taylor series expansion of the function about the mean values of the random variables and neglecting the higher order terms. If the Taylor series is truncated after its linear terms, the following first-order estimates of mean and variance are obtained:

$$\left. \begin{aligned} \mu_Y &= f(\mu_{x1}, \mu_{x2}, \dots, \mu_{xn}) \\ \sigma_Y^2 &= \{b\}^T [C] \{b\} \end{aligned} \right\} \quad \{A7\}$$

where the vector $\{b\}$ denotes $\frac{\partial Y}{\partial x_i}$ evaluated at the mean values of x_i .

If there is no correlation among the variables, Equation (A7) can be written as:

$$\sigma_Y^2 \approx \sum_{i=1}^n \left(\frac{\partial Y}{\partial x_i} \bigg|_{\mu_i} \right)^2 \sigma_{x_i}^2 \quad \{A8\}$$

Equations (A7) or (A8) are referred to as the first-order, second-moment (FOSM) approximations of mean and variance of Y .

The FOSM approximation only provides estimates of the mean and standard deviation, which are not sufficient by themselves for evaluating the failure probability. To estimate the failure probability, one must assume a probability density function for the safety margin or the safety factor. The first step in estimation of failure probability using any probabilistic method is to decide

on what constitutes unsatisfactory performance or failure. Mathematically, this is achieved by defining a performance function $M(X)$, such that $M(X) \geq 0$ means satisfactory performance and $M(X) < 0$ means unsatisfactory performance or “failure”. X is a vector of basic random variables including resistance parameters, load effects, geometry parameters and model uncertainty.

The Reliability Index defined as

$$\beta = \frac{\mu_M}{\sigma_M} \quad \{A9\}$$

in which μ_M and σ_M are, respectively, the mean and standard deviation of the performance function, is often used as an alternative performance measure to the factor of safety [6,19,20,21].

The Reliability Index is a measure of the margin of safety in “*standard deviation units*” and provides more information about the reliability of a geotechnical design or a geotechnical structure than the factor of safety alone. It is directly related to the probability of failure for a normal distribution and the computational procedures used to evaluate the Reliability Index reveal which parameters contribute most to the uncertainty in the factor of safety. This is useful information that can guide the engineer in further investigations. However, the reliability index estimated using the FOSM approach is not “invariant”. The following results show the reliability indices for different formats of the performance function using the FOSM method.

$M(X)$	β
$R - S$	$\frac{F - 1}{\sqrt{F^2 V_R^2 + V_S^2}}$
$\frac{R}{S} - 1$	$\frac{F - 1}{F \sqrt{V_R^2 + V_S^2}}$
$\ln \frac{R}{S}$	$\frac{\ln F}{\sqrt{V_R^2 + V_S^2}}$

where R and S represent, respectively, the total resisting force and the driving force acting on the slope. V_R and V_S denote the coefficients of variation of the resisting and the loading forces respectively and $F = \frac{\mu_R}{\mu_S}$.

The Reliability Index β is the shortest distance in standard deviation units between the most probable set of random variables (the means) and the most probable set of random variables to cause a “failure” ($M = 0$). What the FOSM method is doing is computing the distance from the mean point to the failure surface in the direction of the gradient at the mean point to estimate β . The non-uniqueness of the FOSM method is due to the fact that different functional representations of M may have different mean estimates and different first derivatives. Hasofer and Lind [6] solved the non-uniqueness problem by searching for the “true” minimum distance between the mean point and the failure surface, rather than looking just along the gradient direction.

A2.2. First- and Second-Order Reliability Methods (FORM and SORM)

Hasofer & Lind [6] proposed an invariant definition for the reliability index. The approach is referred to as the First-Order Reliability Method (FORM). The starting point for FORM is the definition of the performance function $M(X)$, where $\mathbf{X}$ is the vector of basic random variables. If the joint probability density function of all random variables $F_X(\mathbf{X})$ is known, then the probability of failure P_f is given by

$$P_f = \int_L F_X(\mathbf{X}) d\mathbf{X} \quad \{A10\}$$

where L is the domain of X where $M(\mathbf{X}) < 0$.

In general, the above integral cannot be solved analytically. In the first-order approximation, the vector of random variables $\mathbf{X}$ is transformed to the standard normal space U , where U is a vector of independent Gaussian variables with zero mean and unit standard deviation, and where $M(U)$ is a linear function. The probability of failure P_f is then:

$$P_f = P[M(U) < 0] \\ \approx P\left[\sum_{i=1}^n \alpha_i U_i - \beta < 0\right] \quad \{A11\}$$

where α_i is the direction cosine of random variable U_i , β is the distance between the origin and the hyperplane $M(U) = 0$ and n is the number of basic random variables X_i .

The vector of the direction cosines of the random variables α_i is called the vector of sensitivity factors, and the distance β is the reliability index. The probability of failure P_f can be estimated from the reliability index using the established equation $P_f = 1 - \Phi(\beta) = \Phi(-\beta)$, where Φ is the cumulative distribution (CDF) of the standard normal variable. The relationship is exact when the limit state surface is planar and the parameters follow normal distributions, and approximate otherwise. The square of the direction cosines or sensitivity factors α_i , whose sum is equal to unity, quantifies in a relative manner the contribution of the uncertainty in each random variable X_i to the total uncertainty.

Low [8] presented a fast methodology for finding the reliability index in the original space using the concept of constrained convex optimisation. His approach is based on the matrix formulation of the Hasofer-Lind reliability index [22, 23]:

$$\beta = \min \sqrt{(\mathbf{X} - \boldsymbol{\mu})^T \mathbf{C}^{-1} (\mathbf{X} - \boldsymbol{\mu})} \quad \{A12\}$$

or, equivalently:

$$\beta = \min \sqrt{\left[\frac{x_i - \mu_i}{\sigma_i}\right]^T [\mathbf{R}]^{-1} \left[\frac{x_i - \mu_i}{\sigma_i}\right]} \quad \{A13\}$$

in which $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_n)$ is the vector of mean values, μ_i, σ_i are the mean value and standard deviations of X_i respectively, $\mathbf{C}$ is the covariance matrix of X and $\mathbf{R}$ is the correlation matrix of $\mathbf{X}$. According to Eq. (A12), the Hasofer-Lind index can be regarded as the minimum distance in units of directional standard deviations from the mean-value point of the random variables to the boundary of the limit state surface. Equation (A13) presented an alternative interpretation of the Hasofer-Lind index based on the perspective of an expanding ellipsoid in the original space of the basic random variables, Fig A2.1.

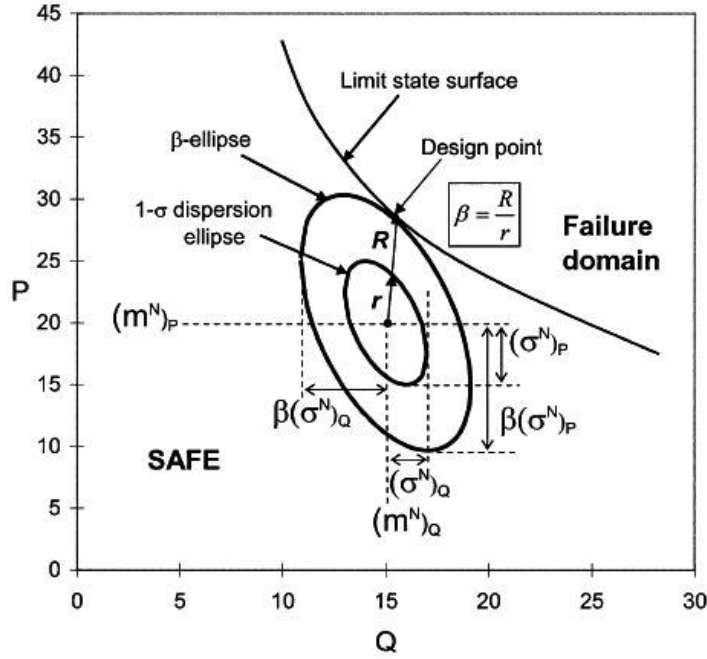


Figure A2.1: Design point and equivalent normal dispersion ellipsoids illustrated in the plane

Notice that Eq. (A13) is only valid for correlated normal distribution of the random variables. For correlated non-normals, Eq. (A13) needs to be rewritten as

$$\beta = \min \sqrt{\left[\frac{x_i - \mu_i^N}{\sigma_i^N} \right]^T [\mathbf{R}]^{-1} \left[\frac{x_i - \mu_i^N}{\sigma_i^N} \right]} \quad \text{[A14]}$$

where μ_i^N and σ_i^N are the equivalent normal mean and equivalent normal standard deviation of random variable x_i , respectively. The values of μ_i^N and σ_i^N can be computed using the Rackwitz-Fiessler two-parameter equivalent normal transformation:

$$\sigma^N = \frac{\phi\{\Phi^{-1}[F(x)]\}}{f(x)} \quad \text{[A15]}$$

$$\mu^N = x - \sigma^N \times \Phi^{-1}[F(x)] \quad \text{[A16]}$$

where x is the original non-normal variant, $\Phi^{-1}[\cdot]$ is the inverse of the standard normal cumulative distribution, $F(x)$ is the original non-normal CDF evaluated at x , $\phi[\cdot]$ is the probabilistic density function of the standard normal distribution and $f(x)$ is the original non-normal probability density ordinate at x . For correlated non-normal, the ellipsoidal perspective and the constrained optimisation approach still apply in the original coordinate system, except that the non-normal distribution are replaced by an equivalent normal hyper-ellipsoid, centred not at the original mean of the non-normal distribution but at the equivalent normal mean μ^N . To facilitate Eq. (A14) in the computation, it is convenient to rewrite Eq. (A14) as:

$$\beta = \min \sqrt{[\mathbf{n}]^T [\mathbf{R}]^{-1} [\mathbf{n}]} \quad \text{[A17]}$$

where $\mathbf{n}$ is a column vector to be optimised. When the value of $\mathbf{n}$ is varied during constrained optimisation, the corresponding value of x_i is calculated from:

$$x_i = F^{-1}[\Phi(n_i)] \quad \text{[A18]}$$

It is commonly accepted that Equation (A14) is in favour of [9] than Equation (A12) because the correlation matrix $\mathbf{R}$ is easier to set up, and conveys the correlation structure more explicitly than the covariance matrix $\mathbf{C}$.

In the second-order reliability method (SORM), the limit state function is defined as in FORM, but the resulting limit state function is approximated by a second order function [24]. However, for many different applications, it has been shown that the probabilities of failure obtained with SORM have been very close to those obtained with FORM [25].

In summary the FORM/SORM approximation involves:

- 1) Transforming a general random vector into a standard Gaussian vector,
- 2) Locating the point of maximum probability density (most likely failure point, Design Point, or simply β -point) within the failure domain, and
- 3) Estimating the probability of failure as $P_f \approx \Phi(-\beta)$.

A2.3. The Random Finite Element Method (RFEM)

The RFEM was pioneered by Griffiths, Fenton and co-workers [26, 27] in the early 1990s since when it has been applied to a wide range of geotechnical applications involving highly variable soils (e.g., steady seepage, settlement, bearing capacity, slope stability, mine pillar stability). This approach involves the generation of random fields of soil/rock properties with properly controlled spatial statistics (mean, standard deviation and spatial correlation). This spatial distribution of properties is then mapped onto quite refined finite element meshes taking proper account of local averaging over each element. A finite element analysis is then performed relating to the particular application under investigation (e.g., thermal diffusivity, thermal expansion, strength analysis).

Monte Carlo simulations then follow, in which the finite element analysis is repeated numerous times (generally in excess of 2000 simulation before a stable trend is observed), each with the same underlying statistics of the soil/rock properties, but different spatial locations of the soil/rock properties across the finite element mesh. Each new random field generation and the subsequent finite element analysis is termed a "realization". By generating a sufficient number of realizations, output quantities of interest after a certain length of time, can be assimilated and statistically analyzed to produce estimates of probability density functions. Supposing $\mathbf{u} = \{u_1, u_2, \dots, u_n\}$ is the vector of response parameters of n realisations. Then the reliability of the response can be computed as

$$\beta = \frac{\mu(\mathbf{u})}{\sigma(\mathbf{u})} \quad \{A19\}$$

The particular benefits of RFEM however, as opposed to FOSM, FORM or other semi-analytical method of probabilistic analysis in geotechnics, can be summarized as follows:

- While the "First order" method can account for local averaging, this must be done a priori by assuming some local averaging region. The RFEM makes no such assumption, since the failure mechanism can "seek out" the path of least resistance through the soil (in failure problems), allowing local averaging to occur naturally.
- The simpler methods lose accuracy in non-linear problems when the "first order" assumption starts to break down. The RFEM with a genuine nonlinear algorithm and Monte-Carlo simulations can cope with both nonlinearity and the higher input variances likely to be observed in geomaterials.
- The RFEM directly models the influence of the spatial correlation length.

- The RFEM/Monte-Carlo approach gives an estimate of the entire PDF of the output quantity of interest, enabling direct estimations of probability.

A2.4. Closing Remarks

This literature review showed that the most unlikely route to pursue is the FOSM due to its non-uniqueness in the reliability structure. At the same time, it showed that FORM and RFEM both have their own advantages and limitations. Even though FORM is fast and probabilistic response of a set of input can be obtained in real time, the method loses its accuracy when encounters with highly non-linear problems. On the other hand, while being robust, the RFEM requires a lot of computational time due to its brute force methodology in obtaining the statistical response.

The main reasons to deploy FORM in the workflow of Fast Probabilistic geomechanical evaluation can be summarized as follows:

1. The Reliability Index (RI) computed from FORM method is invariant
2. The Design Point (DP) can be computed using optimization routines available in Excel. This allows its readily implementation. Modification of limit state function can be done easily.
3. It has a meaningful geometric representation, see Fig. A2.1, in comparison to other probabilistic method. This representation can provide engineering insight into the system being analysed.

Appendix 3. Software Selection



CONTAINMENT: Probabilistic Geomechanics

Decision on software
options and short term
deliverables

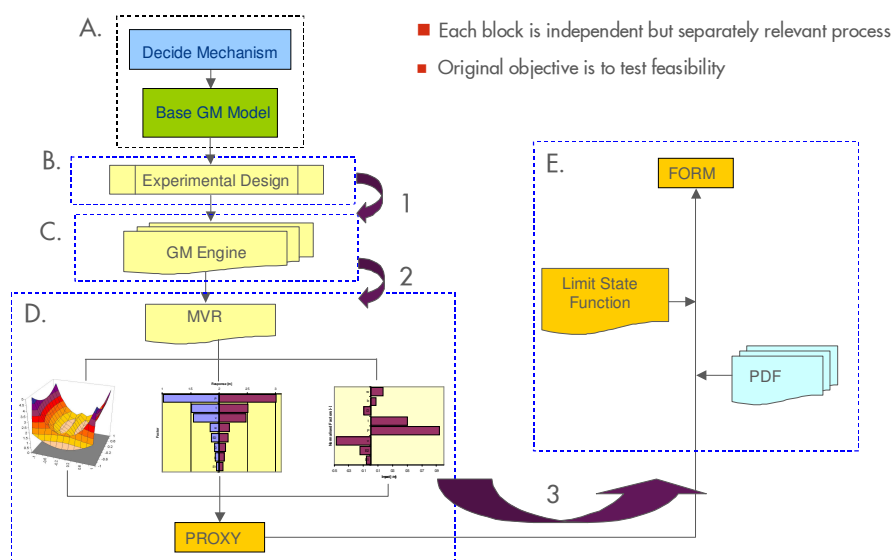
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1

GM RELIABILITY WORKFLOW



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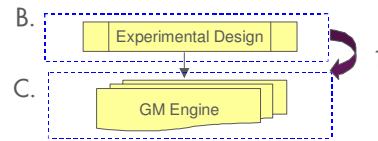
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2

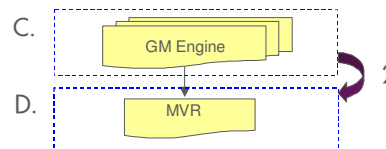
Critical Path 1

- DoE only spits out the table
- GM models with changed state variables still needs to be prepared
- There is no automation from B to C



Critical Path 2

- Output from GM Engine to MVR depends on observation requirements
- Output flexibility from GM Engine is key
- Not aware of this flexibility currently



GM RELIABILITY ANALYSIS SOFTWARE OPTIONS

	B. DOE	C. GM Engine		D. MVR	E. RELIABILITY	
		QB	Geertsma		FORM	MC
Excel GNU License	✓			✓		
Excel			✓ (WIP)	✓	✓	
Python		✓				✓
Prob2B					✓	✓

- What software are we using?
- Short/long term deliverables?
- What time/cost is available?

PROs- AND CONs Excel GNU Software

	B. DOE	C. GM Engine		D. MVR	E. RELIABILITY	
		QB	Geertsma		FORM	MC
Excel software GNU	✓			✓		

■ PROS

- Readily available
- Easy to learn/use
- Pre/Postprocessing
- Multiple analyses
- No cost involve

■ CONS

- Max. 15 state variables
- It "may not" work on Shell Vista

PROs and CONs

	C. Geertsma	D. MVR	E. RELIABILITY
			FORM
Excel	✓ (WIP)	✓	✓

■ PROS

- Independent & no cost

■ CONS

- Primitive functionalities
- No postprocessing
- Tedious multiple analyses

■ PROS

- Independent & no cost

■ CONS

- Primitive functionalities
- No postprocessing
- Tedious rebuild for different analyses

PROs AND CONs

	C. GM Engine		E. RELIABILITY	
	QB	Geertsma	FORM	MC
Python	✓			✓

■ <u>PROS</u> ■ Independent & no cost ■ <u>CONS</u> ■ Learning time/resources
--

■ <u>PROS</u> ■ Independent & no cost ■ <u>CONS</u> ■ Random MC ■ Learning time/resources

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GM RELIABILITY ANALYSIS SOFTWARE OPTIONS

	E. RELIABILITY	
	FORM	MC
Prob2B	✓	✓

- PROS
 - TNO Commercial software
 - Full functions for probabilistic
 - Easy to learn/use
 - Interface with Excel
 - Multiple analyses
 - No maintenance

- CONS
 - Cost \$3500 per license

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Year End deliverables

	B. DOE	C. GM Engine		D. MVR	E. RELIABILITY	
		QB	Geertsma		FORM	MC
Excel GNU License	✓			✓		
Excel			✓ (WIP)	✓	✓	
Python		✓				✓
Prob2B					✓	✓

- Most promising option for B, C and D – Excel
- Most promising option for E – Prob2B. (Need a decision)

Longer Term

- Decision on DOE/MVR/Probability platform
- Translator For Critical Path 1, 2, 3
- Constrained Monte-Carlo

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